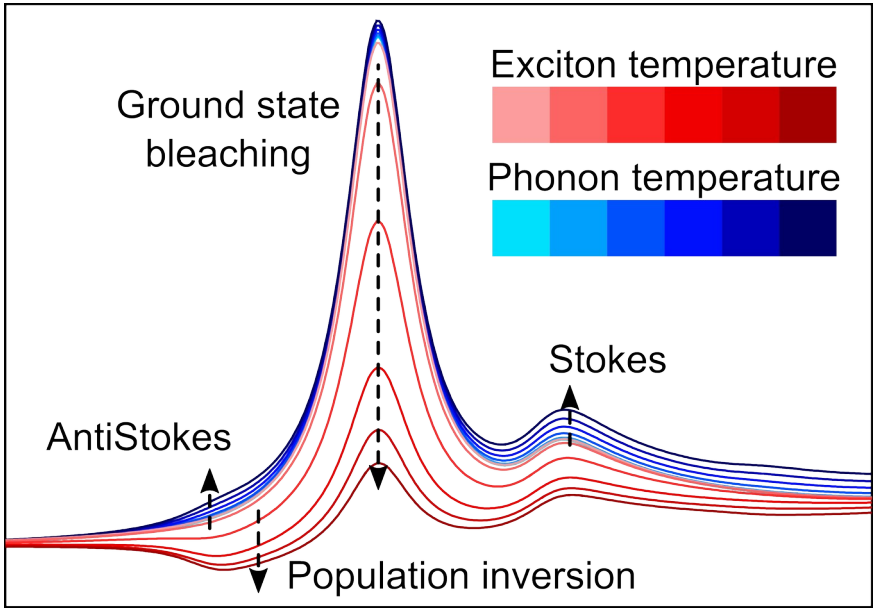
NANYANG TECHNOLOGICAL UNIVERSITY

Phonon-mediated excitonic population inversion in (6,5) single walled carbon nanotubes

Stefano Dal Forno, Marco Battiato



Motivation

Carbon nanotubes (CNTs):

- quasi-1D systems, can be semiconducting or metallic.
- can be used for opto-electronic devices;
- for chemical or gas sensors;
- strong and flexible conducting devices;
- for enhancing thermal, mechanical and electronic properties.
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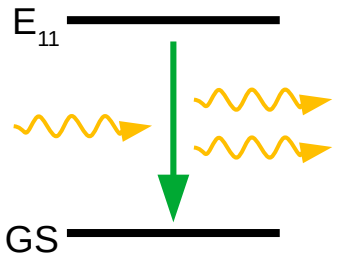
We want to describe the conditions to obtain a **laser light emission** in CNTs.

- Which conditions guarantee to achieve **population inversion;**
- What are the **energy costs;**
- Which **systems are best.**

Stimulated light emission in CNTs



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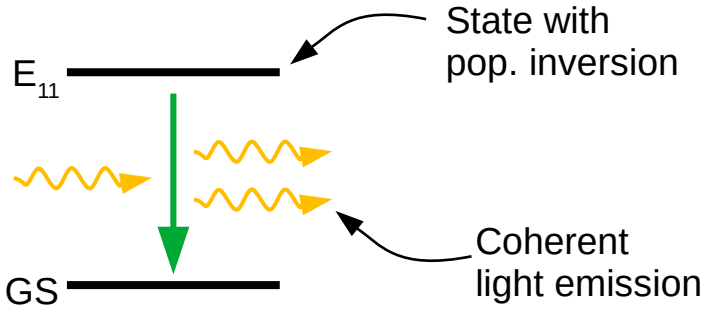

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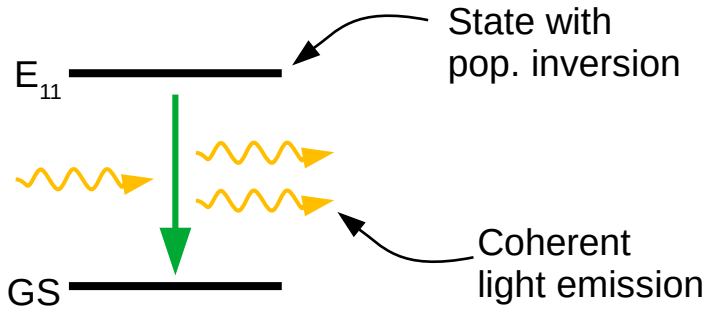

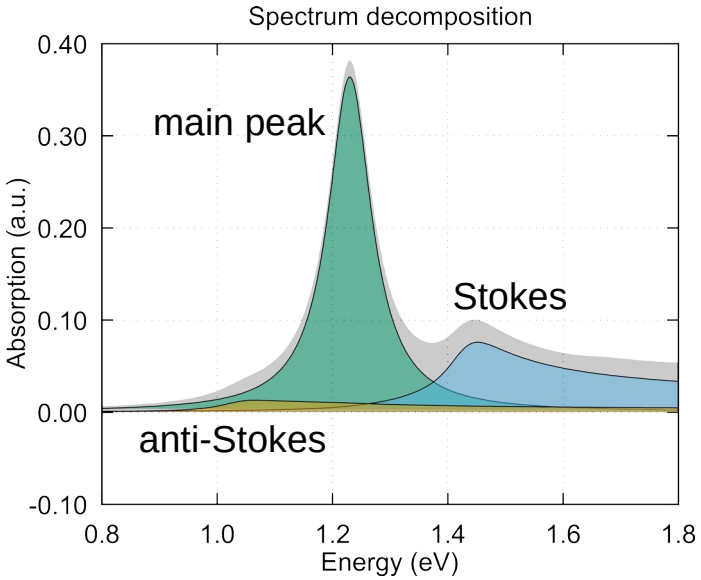

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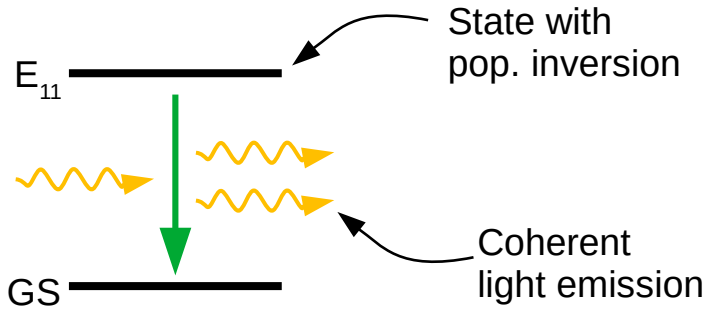

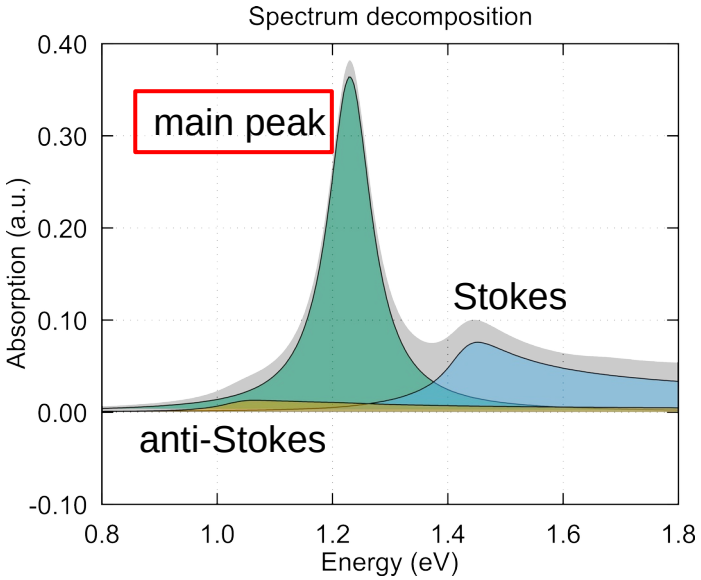

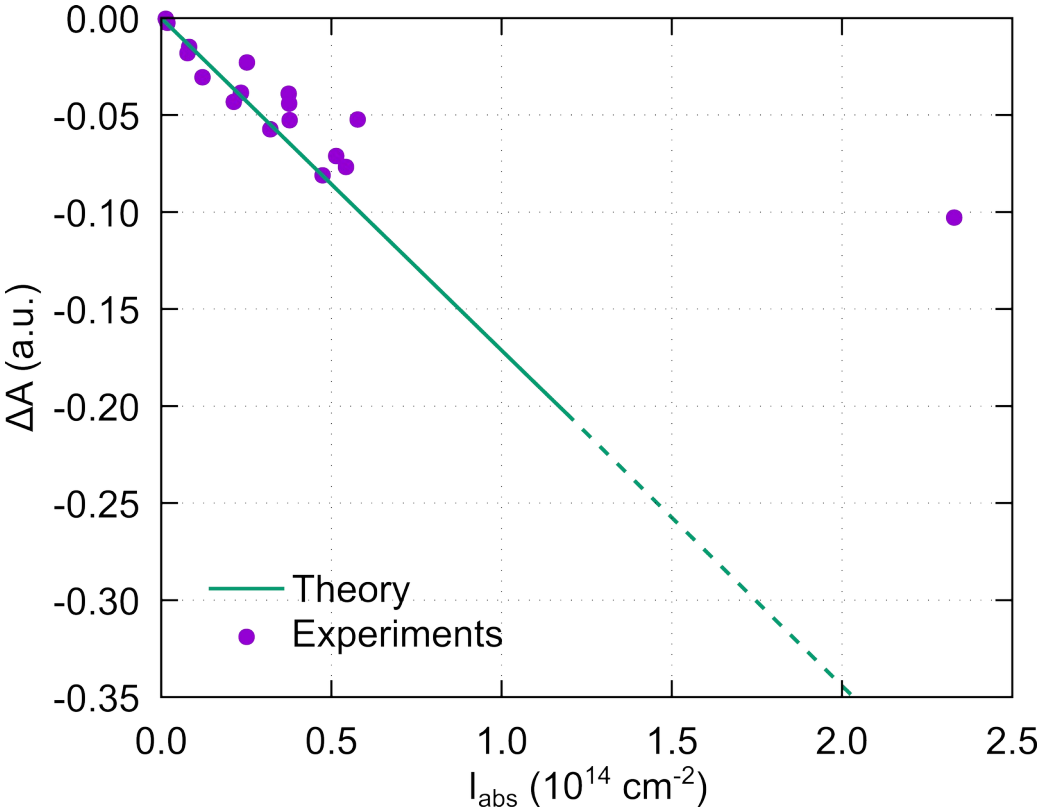

Larry Lueer, et al. *"Size and mobility of excitons in (6, 5) carbon nanotubes"* Nature Physics, 5(1):54–58, January 2009

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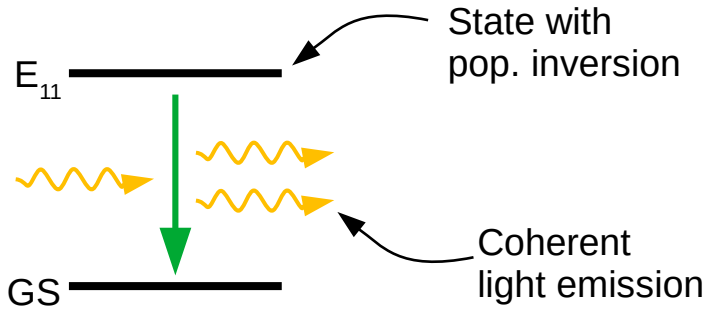

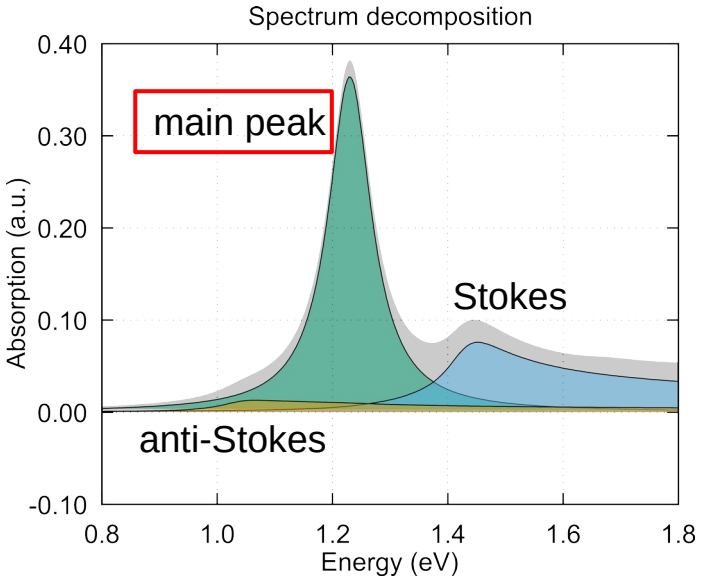

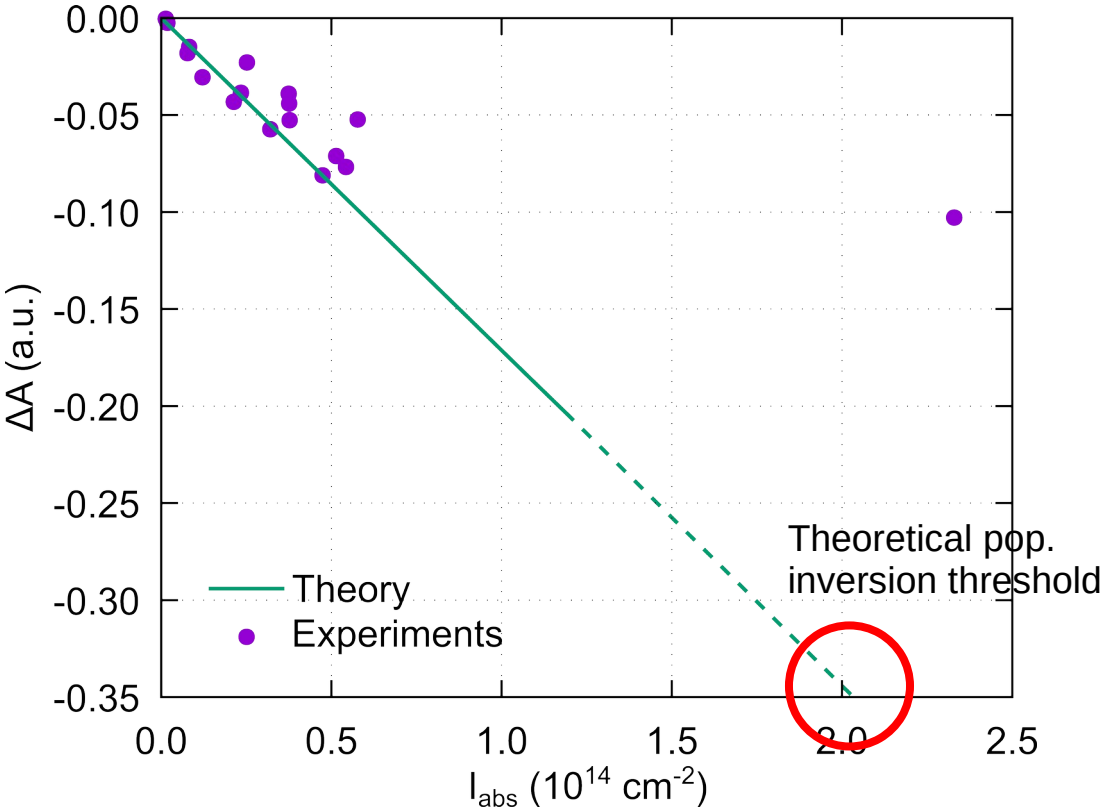

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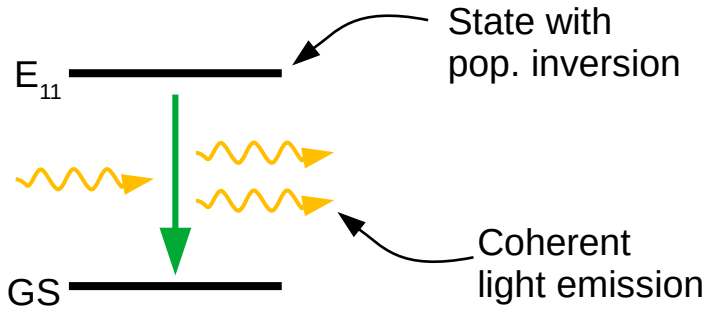

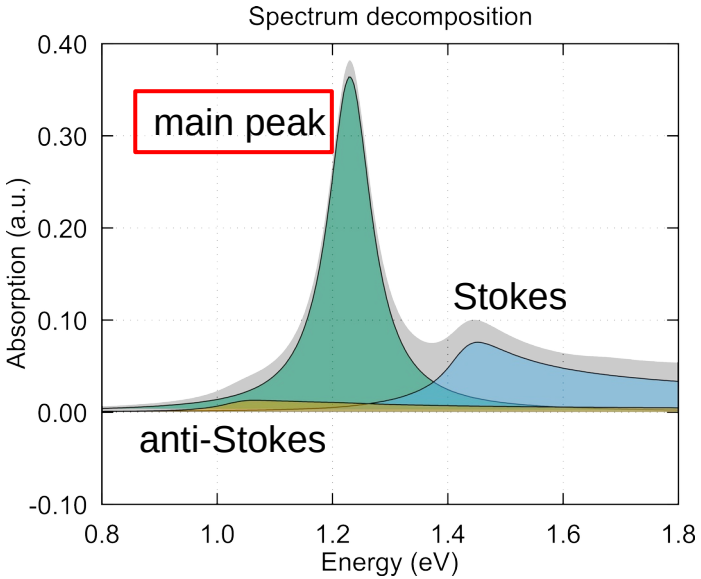

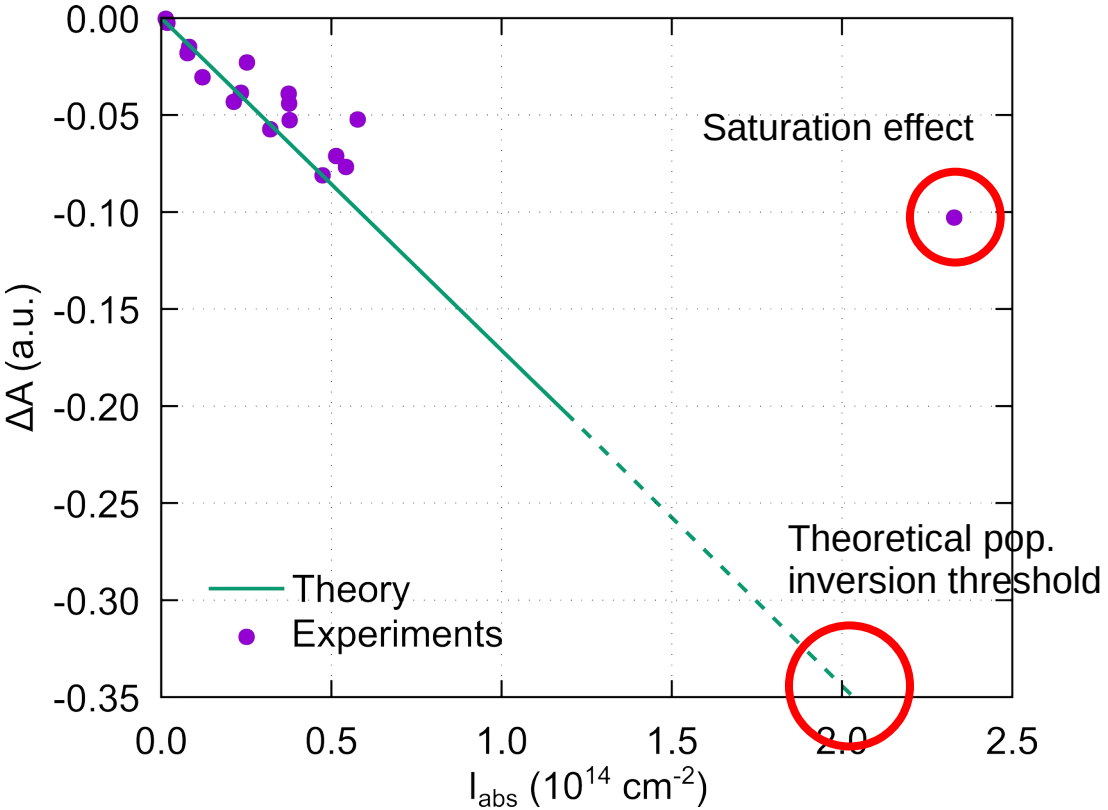

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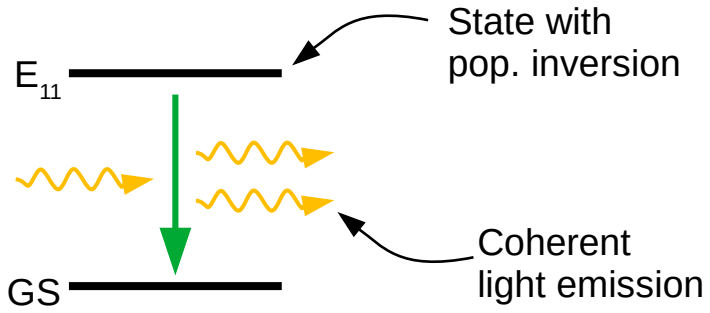

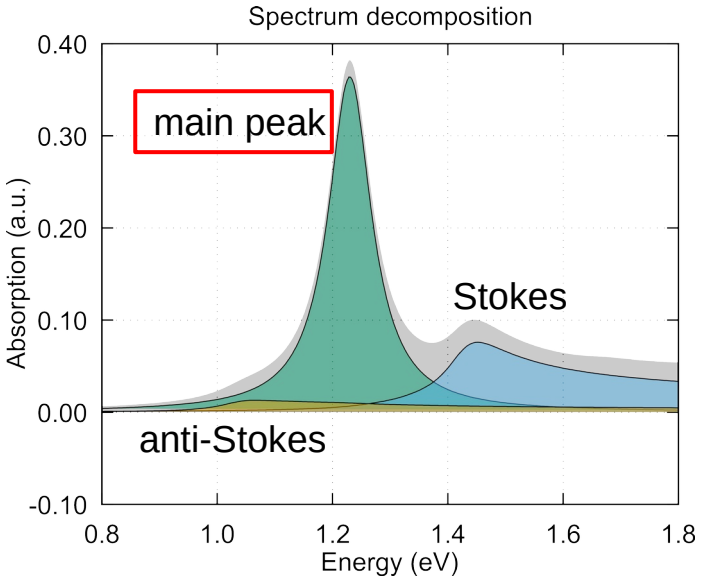

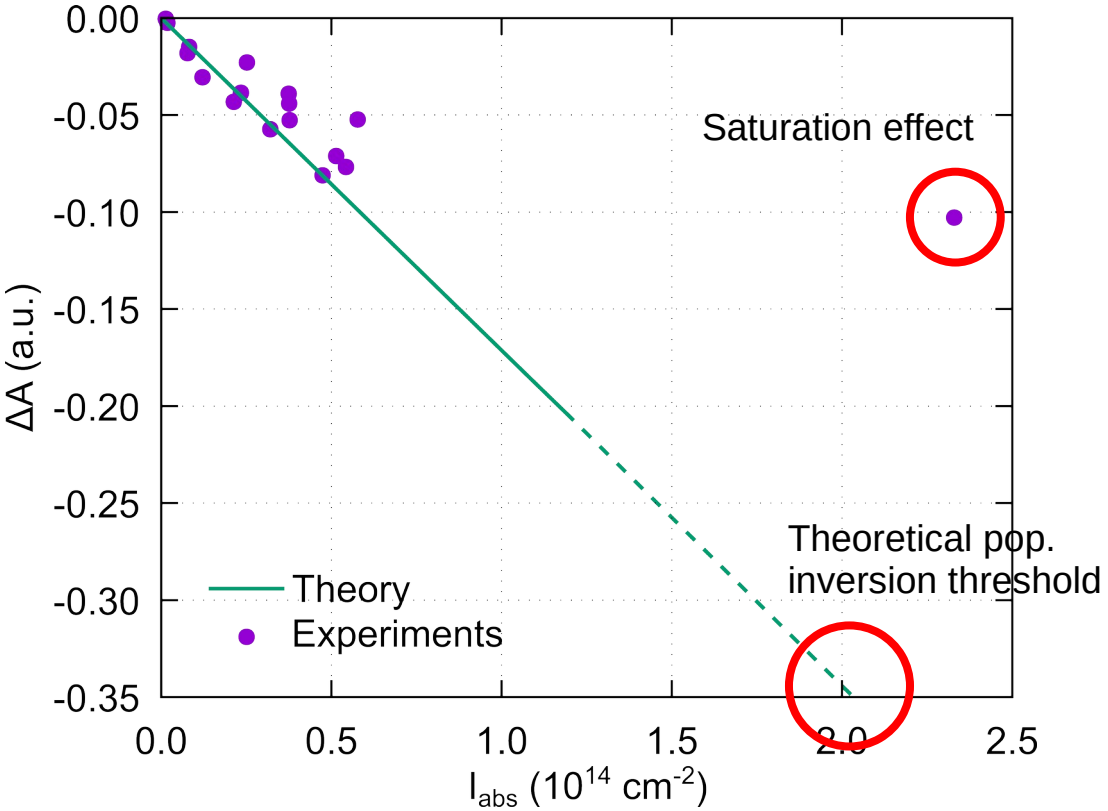

Pop. inversion of the main excitonic transition is unfeasible!

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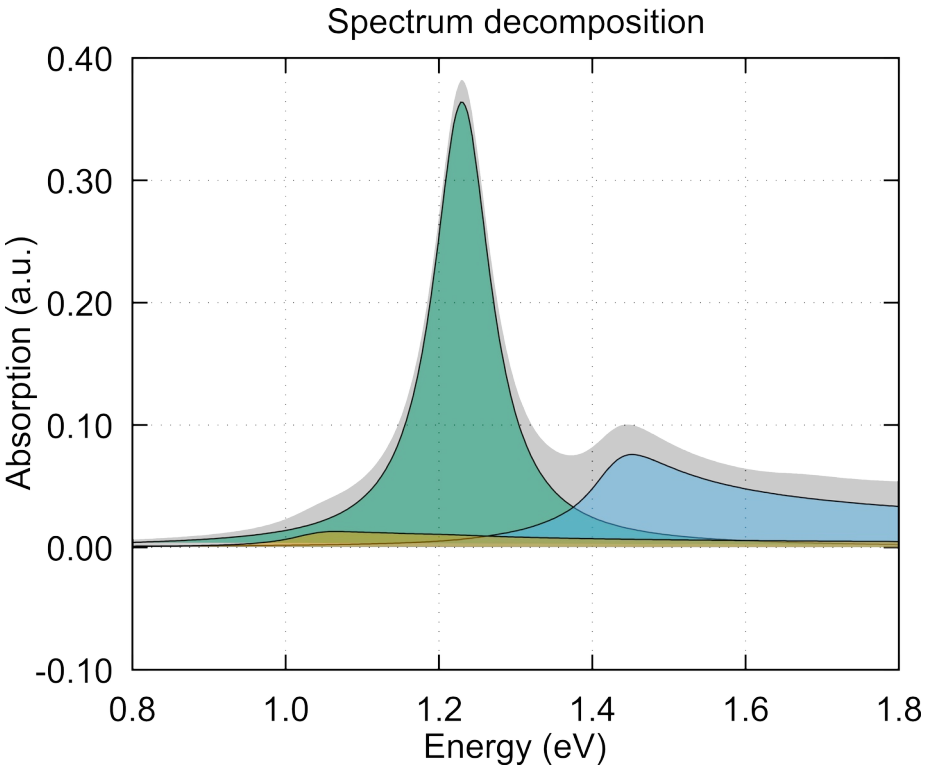
Stimulated light emission in CNTs

What if we focus on the **phonon side band transitions**?



The model

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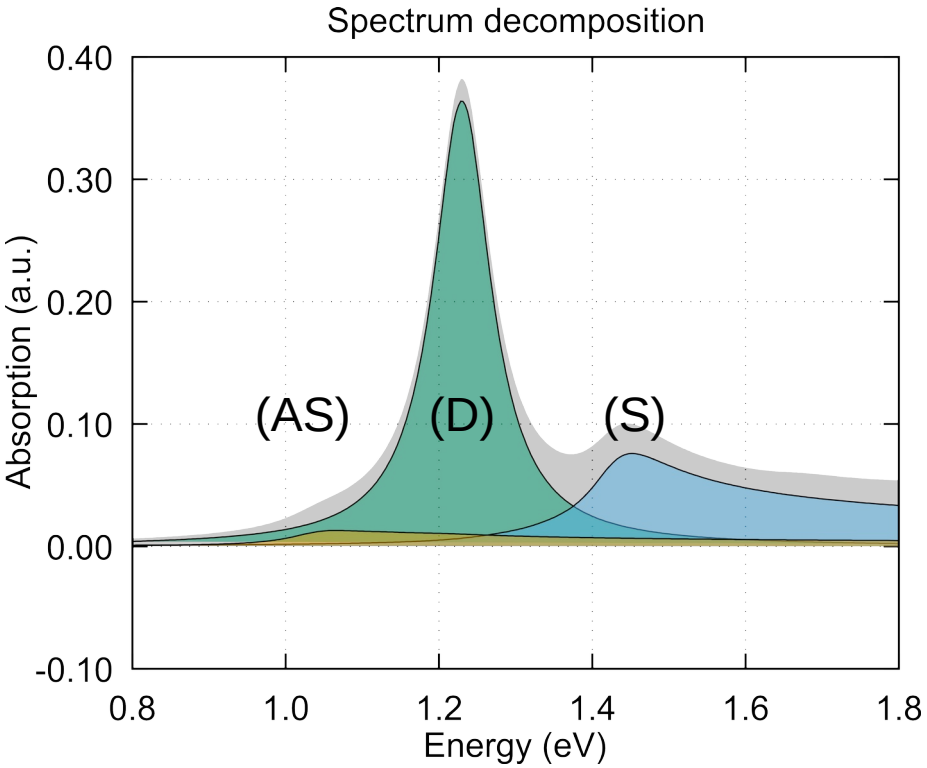
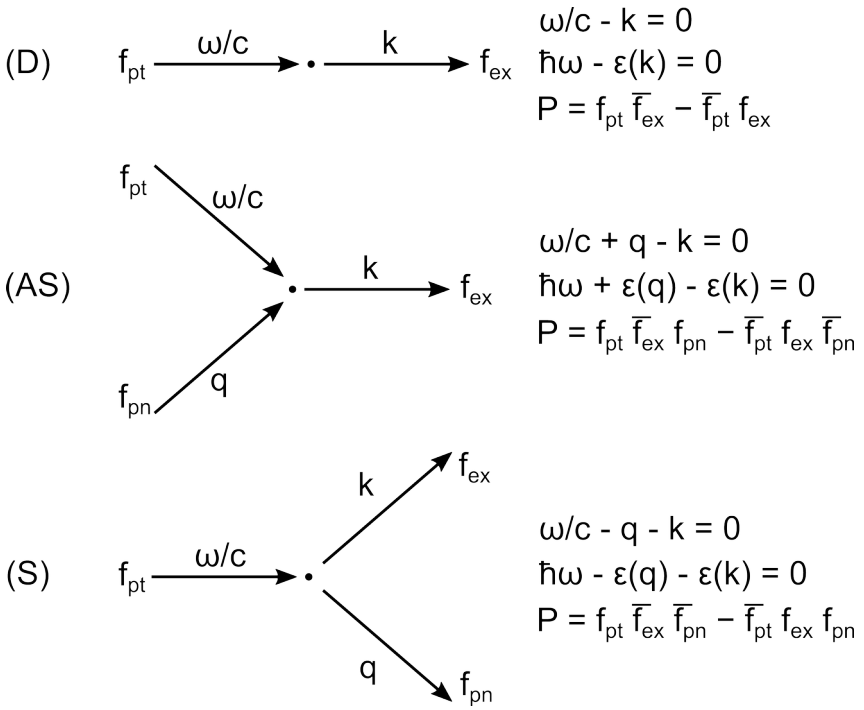
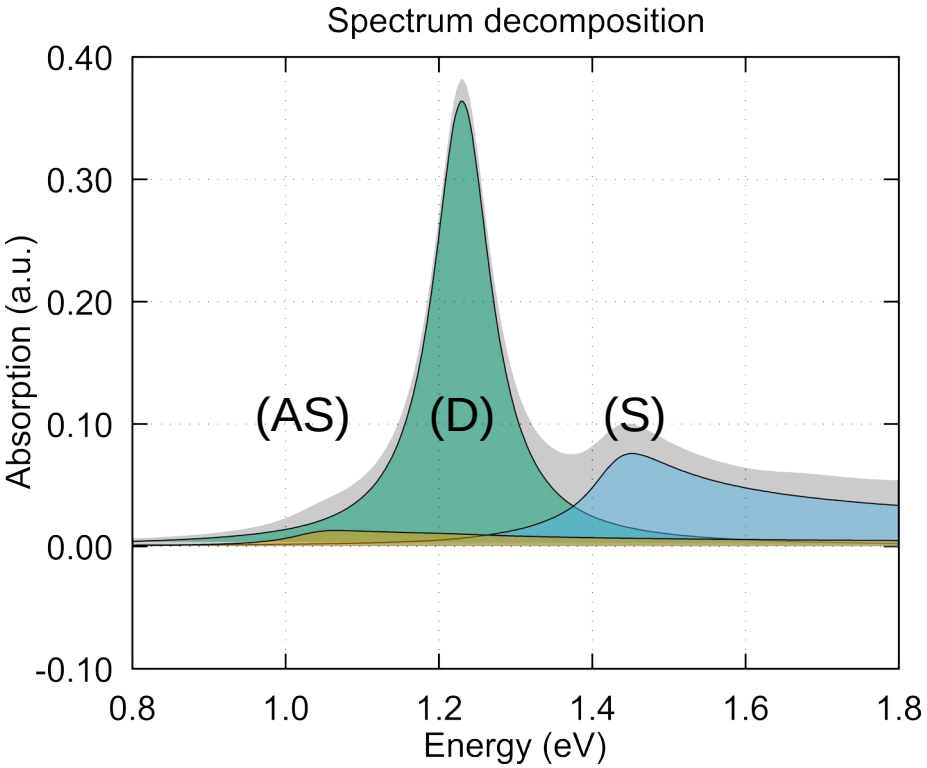




Stefano Dal Forno, et al. *"Origin of the background absorption in carbon nanotubes: Phonon-assisted excitonic continuum"* Carbon, 186:465–474, January 2022

The model

We will use the Boltzmann equation



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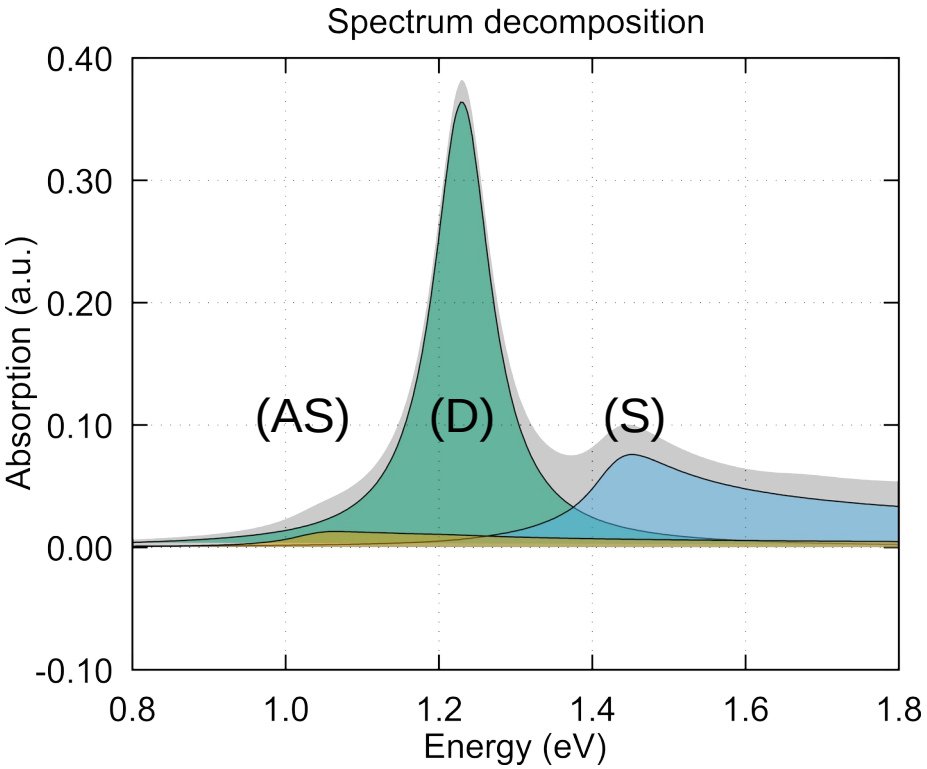

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$$\mathcal{I}_\alpha = \hbar\int \lambda_\alpha(\omega)d\omega,$$

The above are:

- the probability rate of absorbing (or emitting) a photon;
- The total power absorbed by the scattering process.

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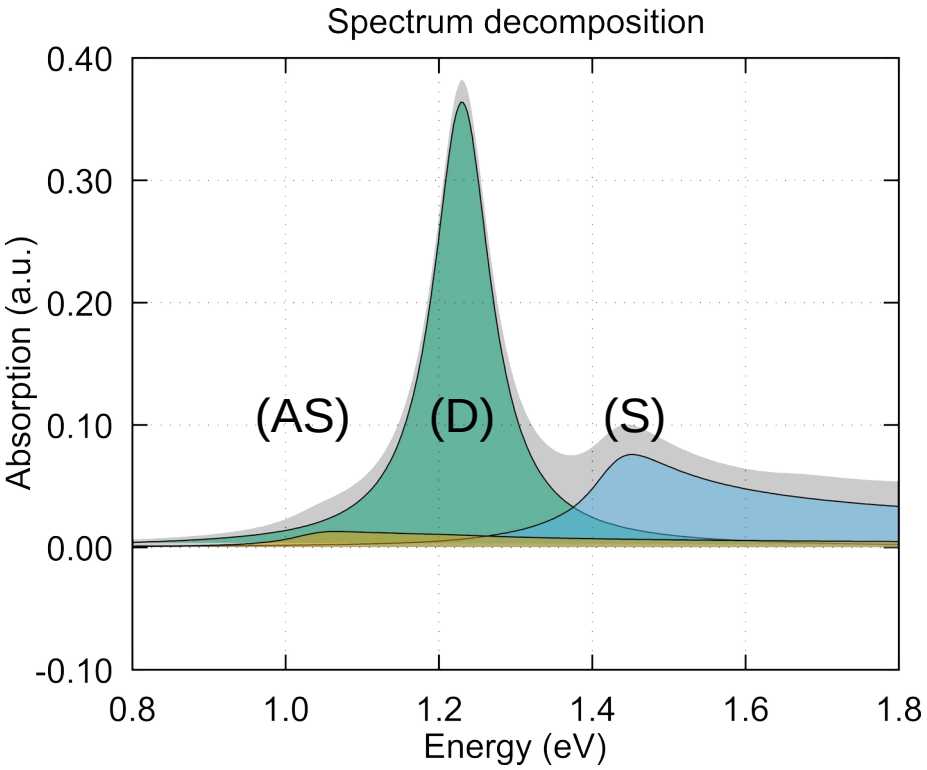

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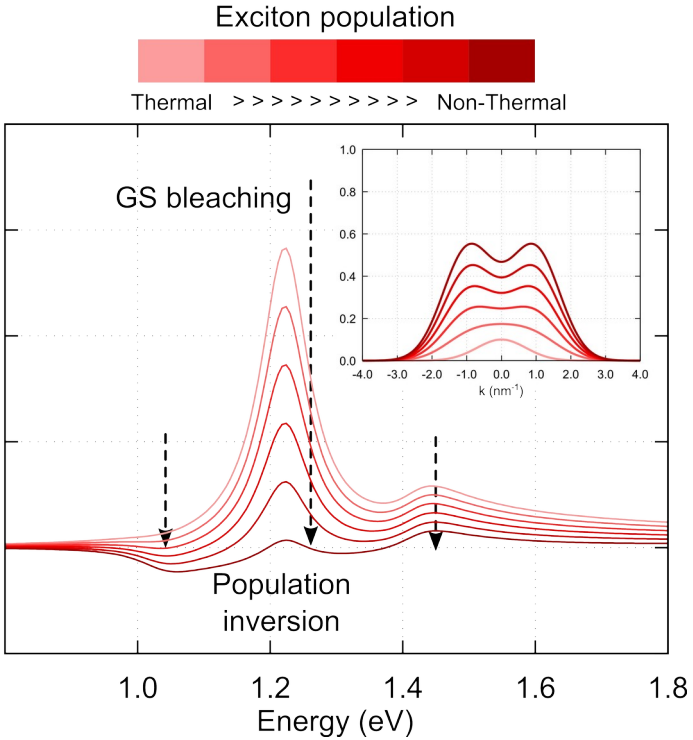
AIM: which conditions on the exciton population guarantee light emission?

Results

Probability rates and power absorbed, as function of the exciton population



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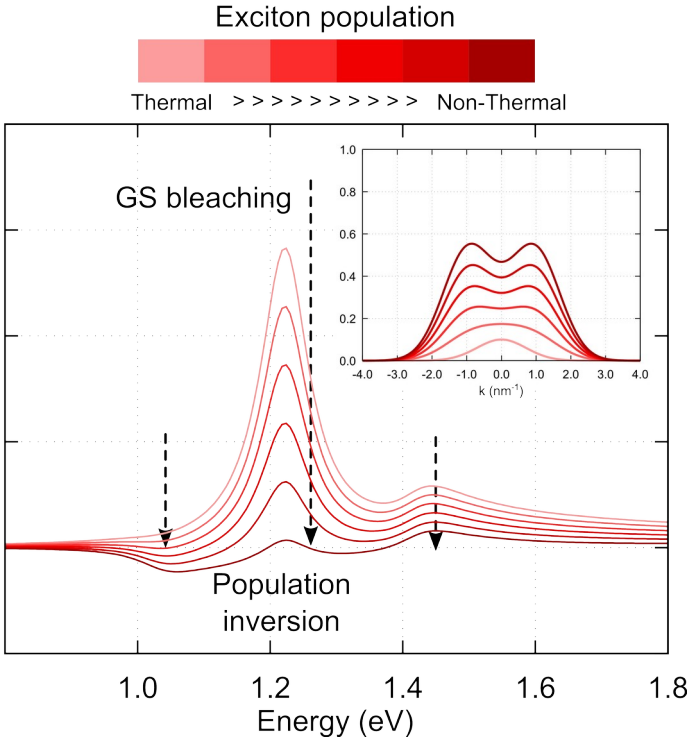

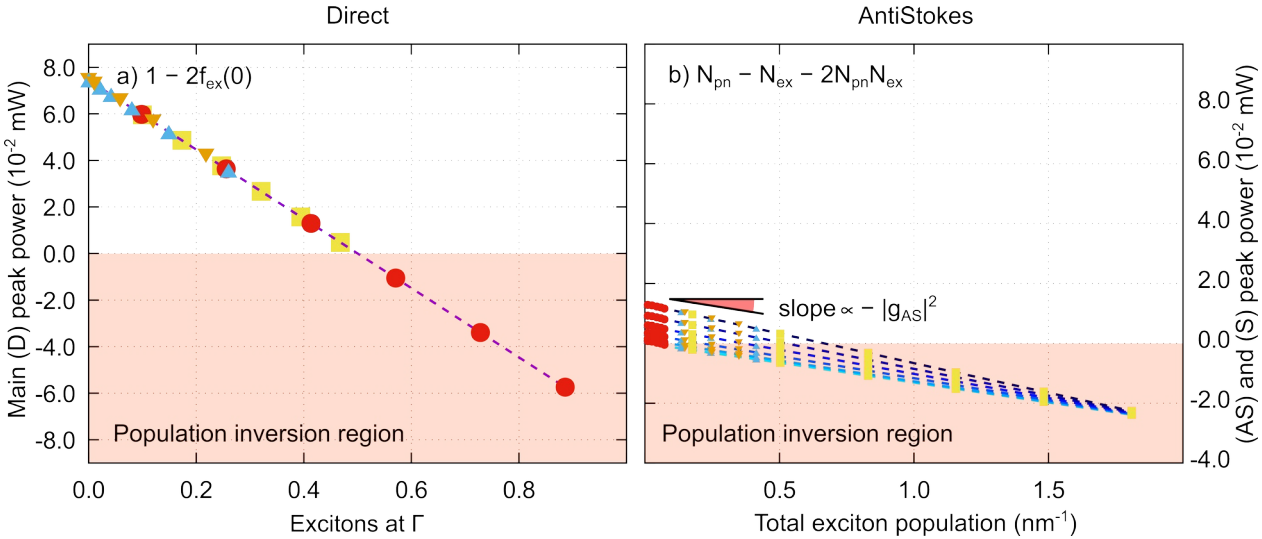

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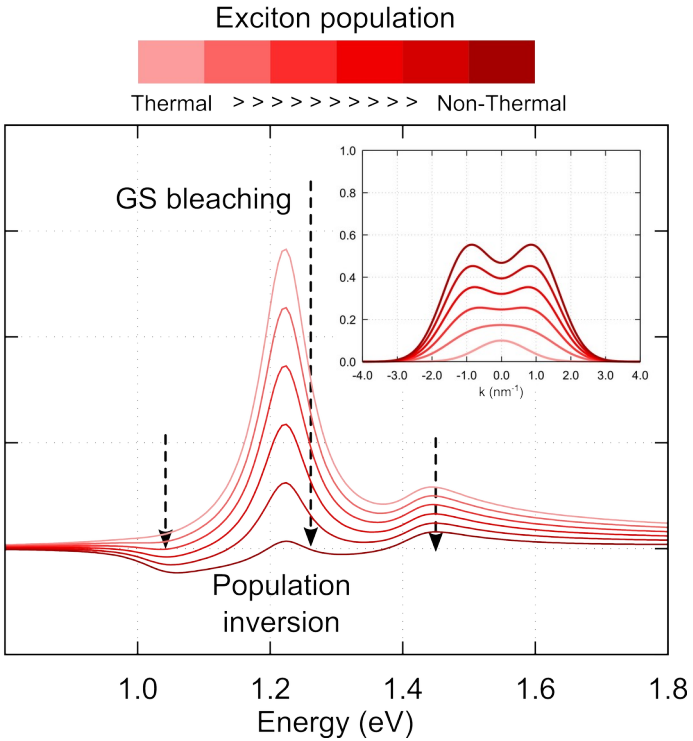

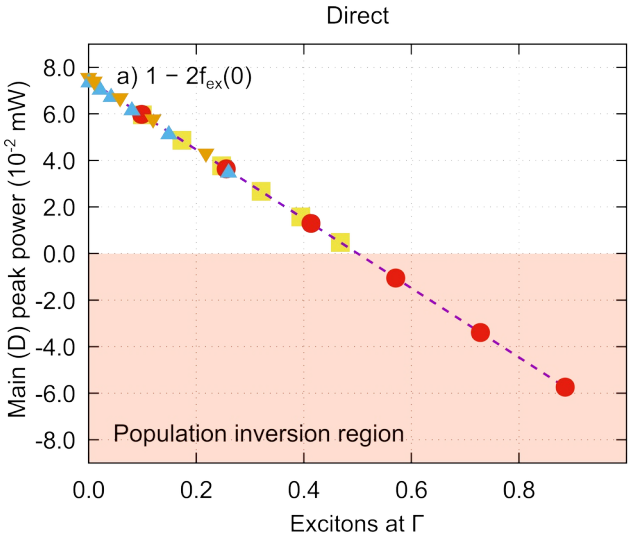

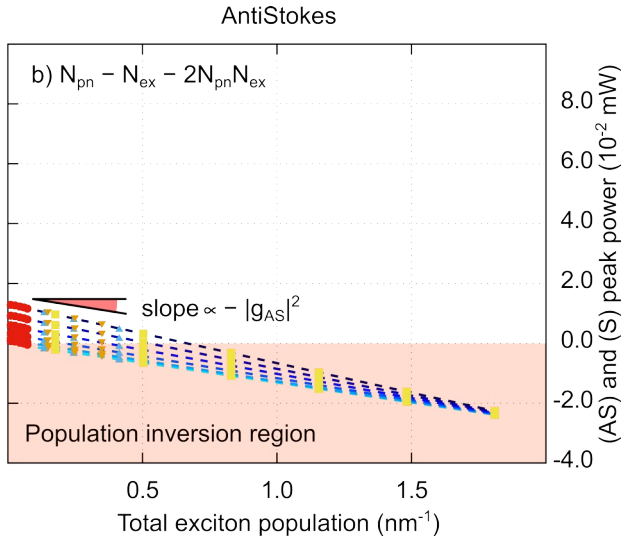

The main D peak intensity only depends on the excited excitons at the Gamma point.
Direct **pop. inversion**:
1) Happens at exactly half occupancy;
2) Is difficult to achieve because of the high laser fluence required.

Stimulated emission mediated by AS phonons:
1) Is stronger at lower temperatures;
2) Depends on the exciton-phonon coupling strength;
3) Can be achieved with very low power.

Conclusion and aknowledgements

- Excitonic population inversion is achievable with the help of Anti-Stokes process;
- It scales linearly with the exciton-phonon coupling strength;
- Better interpretation of pump-probe experimental results;
- Our model is capable of simulate truly out of equilibrium systems.

Special thanks:

Quantum BE solver

github.com/MarcoBattiato/TORTOISE

Carbon NanoTube Tool

github.com/t3n0/cntt

Stefano Dal Forno

Prof. Marco Battiato

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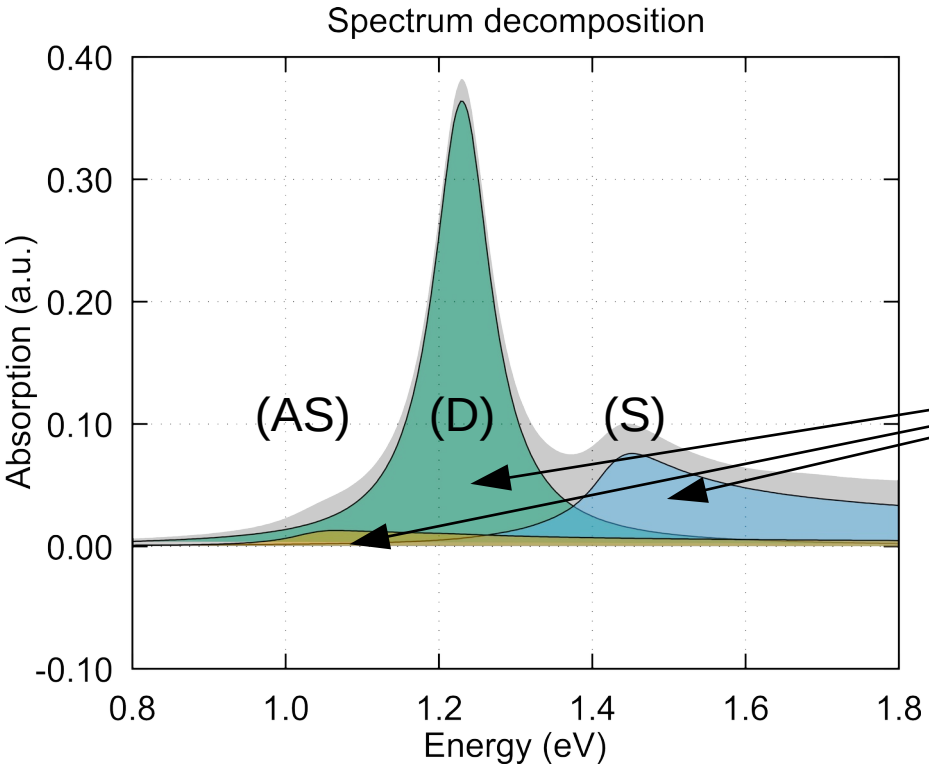

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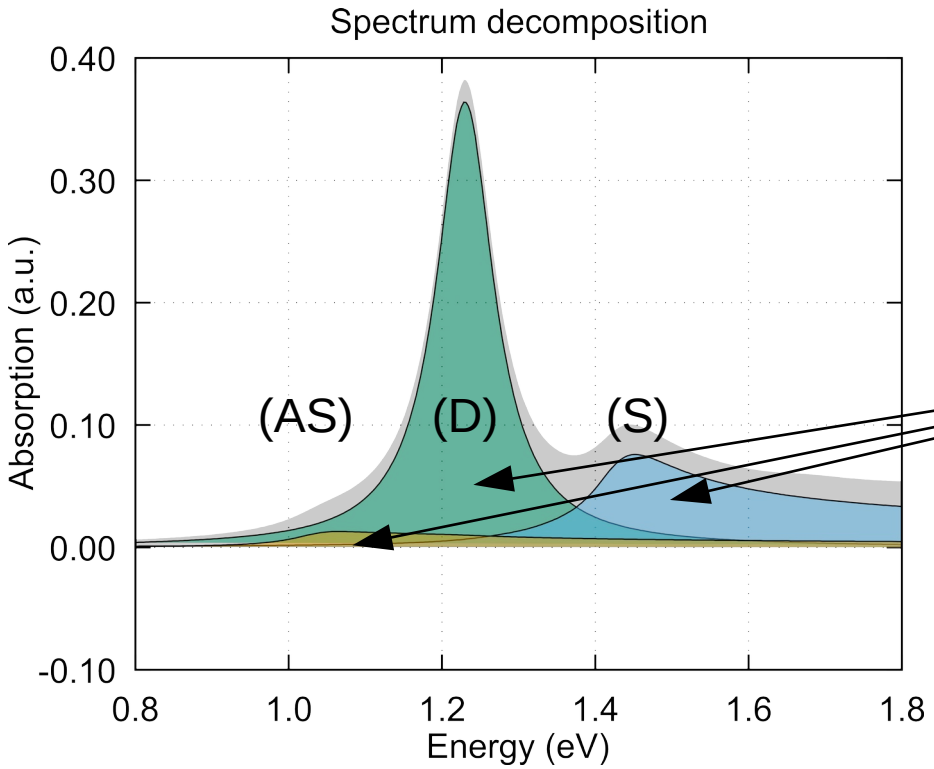

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	Exciton population factors
D	$1 - 2f_{ex}(0)$
AS	$N_{pn} - N_{ex} - 2N_{ex}N_{pn}$
S	$1 + N_{pn} - N_{ex} - 2N_{ex}N_{pn}$

Results

Absorption spectrum changes as function of the exciton and phonon populations

- Photons and phonons are modelled as Bosons;
- Excitons are modelled as Fermions.

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- Excitons are modelled using 4 representative test populations:
  1) Thermal excitons;
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  3) Laser excited excitons;
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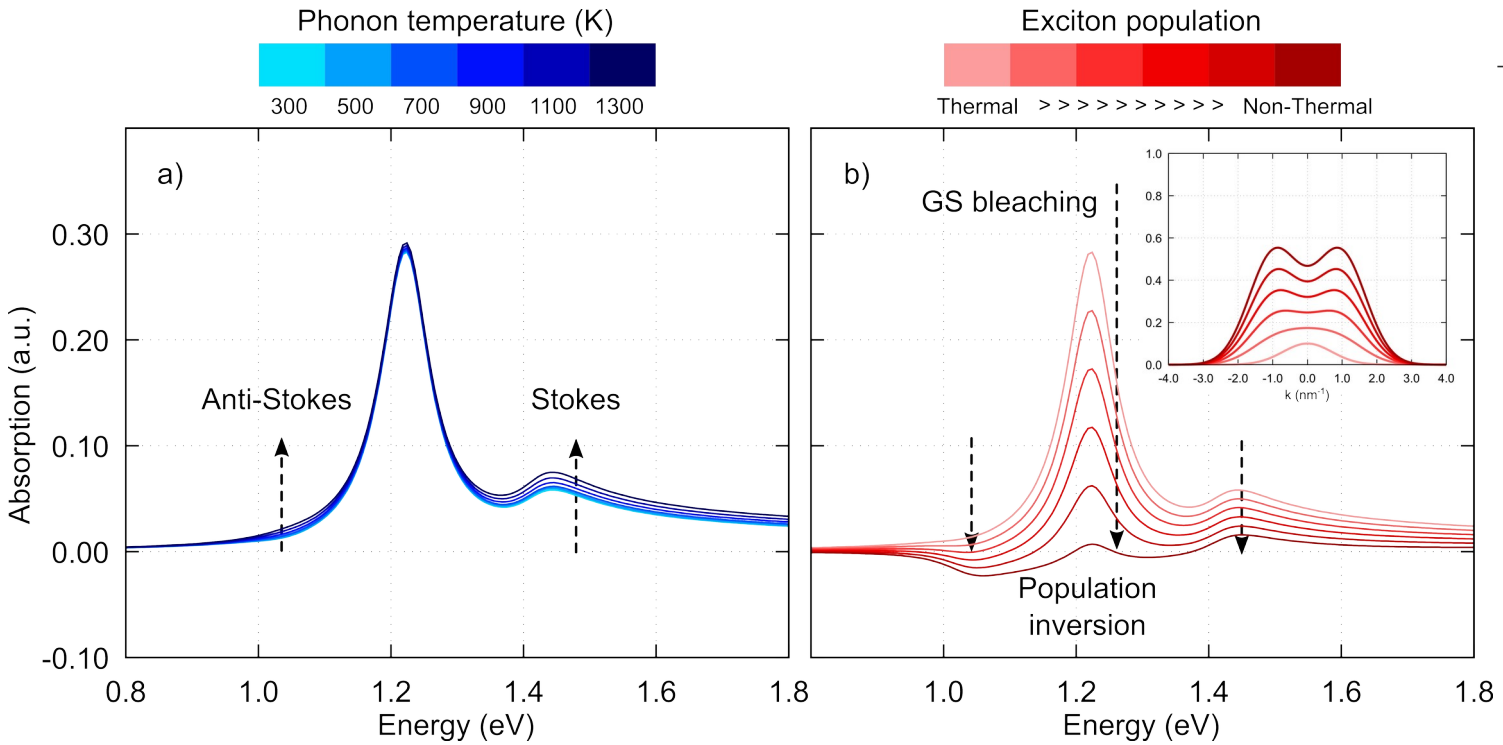

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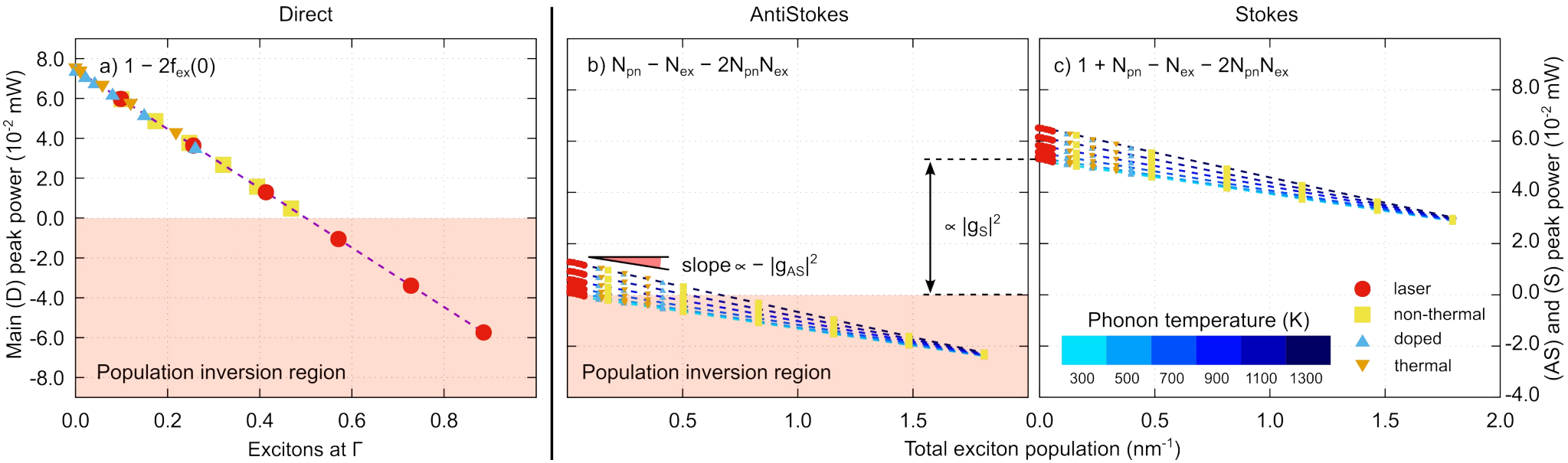
- AS and S peaks increase with phonon temperature
- Main D peak is not affected

- All peaks reduce the intensity with exciton population
- Coupling with AS phonons produce **population inversion**

Dr. Stefano Dal Forno

Results

Peak intensities as function of exciton population



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Pop. inversion mediated by AS phonons:

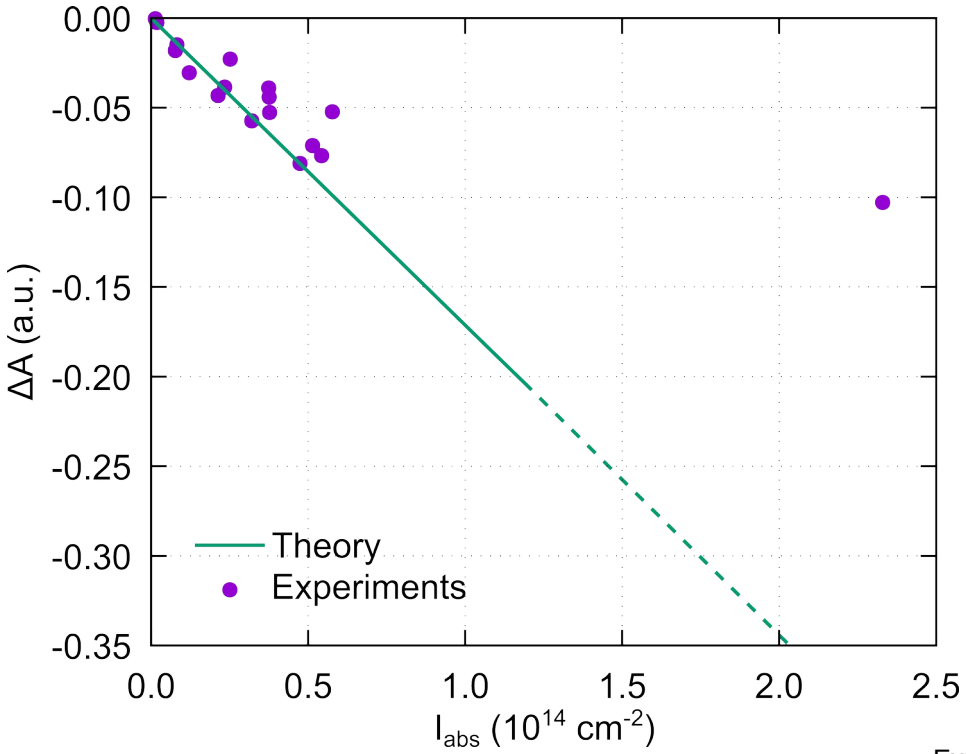
1) Is stronger at lower temperatures;
2) Depends on the exciton-phonon coupling strength.

The +1 addend in the pop. **phase factors forbids** a mediated pop. inversion by S phonons:

1) The S – AS intensity difference is prop. to the coupling strength;
2) The AS and S peaks are proportional to the total exciton population (not just at Gamma point)

Results

Comparison with pump-probe experiments



- Excellent agreement at low exciton densities;
- At high density, experiments observe a saturation effect (cf. isolated point on the right)

- I_{abs} and ΔA are different quantities than excitons at Gamma and the peak intensity

Experimental data from Larry Lueer, et al. *"Size and mobility of excitons in (6, 5) carbon nanotubes"* Nature Physics, 5(1):54–58, January 2009

The quantum Boltzmann equation

To study the dynamics and the interactions between charge carriers, phonons and excitons in CNTs, we invoke the quantum Boltzmann equation:

$$\frac{\partial f_n}{\partial t} + \mathbf{v}_n \cdot \frac{\partial f_n}{\partial \mathbf{r}} - \frac{e}{\hbar}\left[\mathbf{E} + \frac{\mathbf{v}_n}{c} \times \mathbf{B}\right] \cdot \frac{\partial f_n}{\partial \mathbf{k}} = \left(\frac{\partial f_n}{\partial t}\right)_{coll}$$

$$\frac{\partial f_n}{\partial t} = \frac{\partial f_n}{\partial t}(t, \mathbf{r}, \mathbf{k}) \qquad \mathbf{v}_n(\mathbf{k}) = \frac{1}{\hbar}\frac{\partial \epsilon_n(\mathbf{k})}{\partial \mathbf{k}}$$

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To study the dynamics and the interactions between charge carriers, phonons and excitons in CNTs, we invoke the quantum Boltzmann equation:

$$\frac{\partial f_n}{\partial t} + \cancel{\mathbf{v}_n \cdot \frac{\partial f_n}{\partial \mathbf{r}}} - \cancel{\frac{e}{\hbar}\left[\mathbf{E} + \frac{\mathbf{v}_n}{c} \times \mathbf{B}\right] \cdot \frac{\partial f_n}{\partial \mathbf{k}}} = \left(\frac{\partial f_n}{\partial t}\right)_{coll}$$

$$\frac{\partial f_n}{\partial t} = \frac{\partial f_n}{\partial t}(t, \cancel{\mathbf{r}}, \mathbf{k}) \qquad \mathbf{v}_n(\mathbf{k}) = \frac{1}{\hbar}\frac{\partial \epsilon_n(\mathbf{k})}{\partial \mathbf{k}}$$

The quantum Boltzmann equation

We are left with:

$$\frac{\partial f_n}{\partial t} = \left(\frac{\partial f_n}{\partial t}\right)_{coll}$$

The quantum Boltzmann equation

We are left with:

$$\frac{\partial f_n}{\partial t} = \left(\frac{\partial f_n}{\partial t}\right)_{coll}$$

A commonly used approximation is the quantum Fokker-Planck equation:

$$\left(\frac{\partial f_1}{\partial t}\right)_{12-34} = \sum_{\mathbf{G}} \int\int\int_{V_{BZ}^3} d\mathbf{k}_2 d\mathbf{k}_3 d\mathbf{k}_4 \left|g_{34}^{12}\right|^2 \delta_{\mathbf{k}}\, \delta_{\epsilon}\, P_{34}^{12}$$

$$\delta_{\mathbf{k}} = \delta(\mathbf{k}_1 + \mathbf{k}_2 - \mathbf{k}_3 - \mathbf{k}_4 + \mathbf{G})$$

$$\delta_{\epsilon} = \delta(\epsilon_1(\mathbf{k}_1) + \epsilon_2(\mathbf{k}_2) - \epsilon_3(\mathbf{k}_3) - \epsilon_4(\mathbf{k}_4))$$

$$P_{34}^{12} = [(1-f_1)(1-f_2)f_3 f_4 - f_1 f_2 (1-f_3)(1-f_4)]$$

(electron-electron scattering example)

Solving the quantum Fokker-Planck equation

$$\left(\frac{\partial f_1}{\partial t}\right)_{12-34} = \sum_{\mathbf{G}} \int\int\int_{V^3_{BZ}} d\mathbf{k}_2 d\mathbf{k}_3 d\mathbf{k}_4 \left|g^{12}_{34}\right|^2 \delta_{\mathbf{k}}\, \delta_{\epsilon}\, P^{12}_{34}$$

Very difficult because:

- It is a 12 dimensional integral;
- It is a 4th order operator due to the population factor;
- Energy and momentum deltas are difficult to handle;
- Very difficult to converge.

Solving the quantum Fokker-Planck equation

$$\left(\frac{\partial f_1}{\partial t}\right)_{12-34} = \sum_{\mathbf{G}} \int\int\int_{V^3_{BZ}} d\mathbf{k}_2 d\mathbf{k}_3 d\mathbf{k}_4 \left|g^{12}_{34}\right|^2 \delta_{\mathbf{k}}\, \delta_{\epsilon}\, P^{12}_{34}$$

Our approach:

- Project the integrand onto a set of localised basis functions with compact support:
  → The scattering integral become a sparse tensor.
- Basis functions are second order polynomial:
  → Fewer k point needed. Exact inversion of Dirac deltas.
- Analytic inversion of Dirac deltas:
  → The dimensionality of the integral is reduced. Exact energy and momentum conservation.

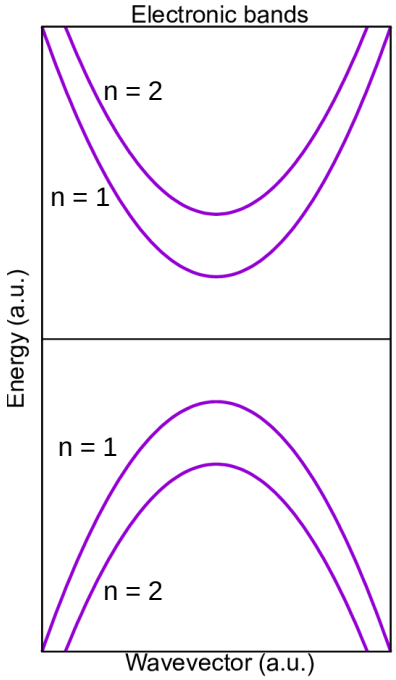
Solving the quantum Fokker-Planck equation

$$\left(\frac{\partial f_1}{\partial t}\right)_{12-34} = \sum_{\mathbf{G}} \int\int\int_{V^3_{BZ}} d\mathbf{k}_2 d\mathbf{k}_3 d\mathbf{k}_4 \left|g^{12}_{34}\right|^2 \delta_{\mathbf{k}}\, \delta_{\epsilon}\, P^{12}_{34}$$

From the quantum Fokker-Plank equation we can compute:

- The static absorption: proportional to the inverse of the photon scattering rate;
- The dynamical absorption, as the populations evolve in time.

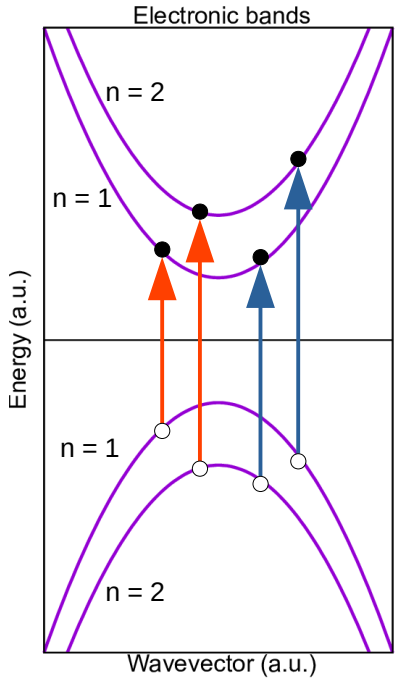
Electronic band structure of (6,5) CNTs



Sketch of the band structure of a semiconducting CNT:

- n is the sub-band index quantum number;

Electronic band structure of (6,5) CNTs



Sketch of the band structure of a semiconducting CNT:

- n is the sub-band index quantum number;

When a optical transition occurs, n is not conserved.

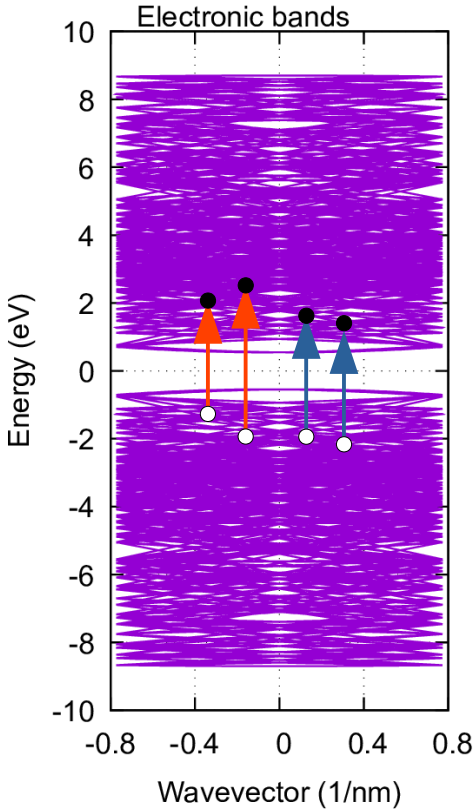
Light polarised parallel to the CNT axis has $\Delta n = 0$.
Light perpendicular to the CNT axis has $\Delta n = \pm 1$.

On the other hand, the change in wavevector is conserved.

Transitions:

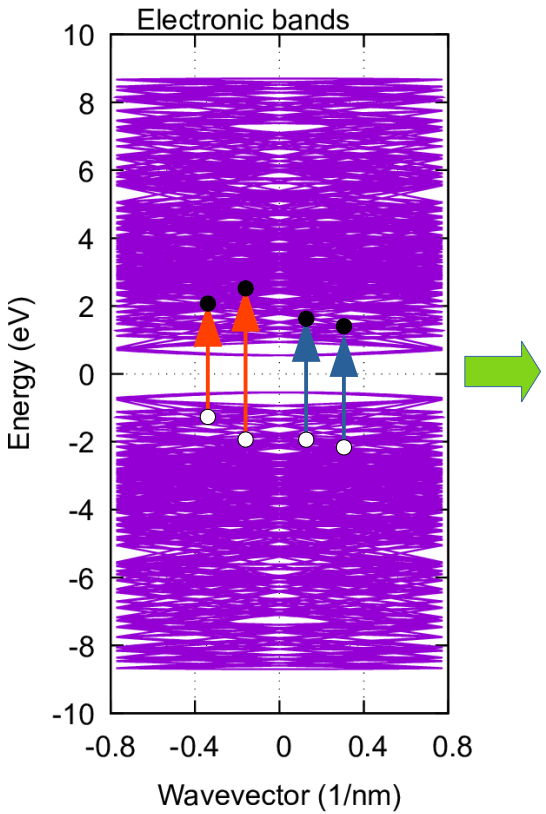
- parallel polarised light
- perpendicular polarised light

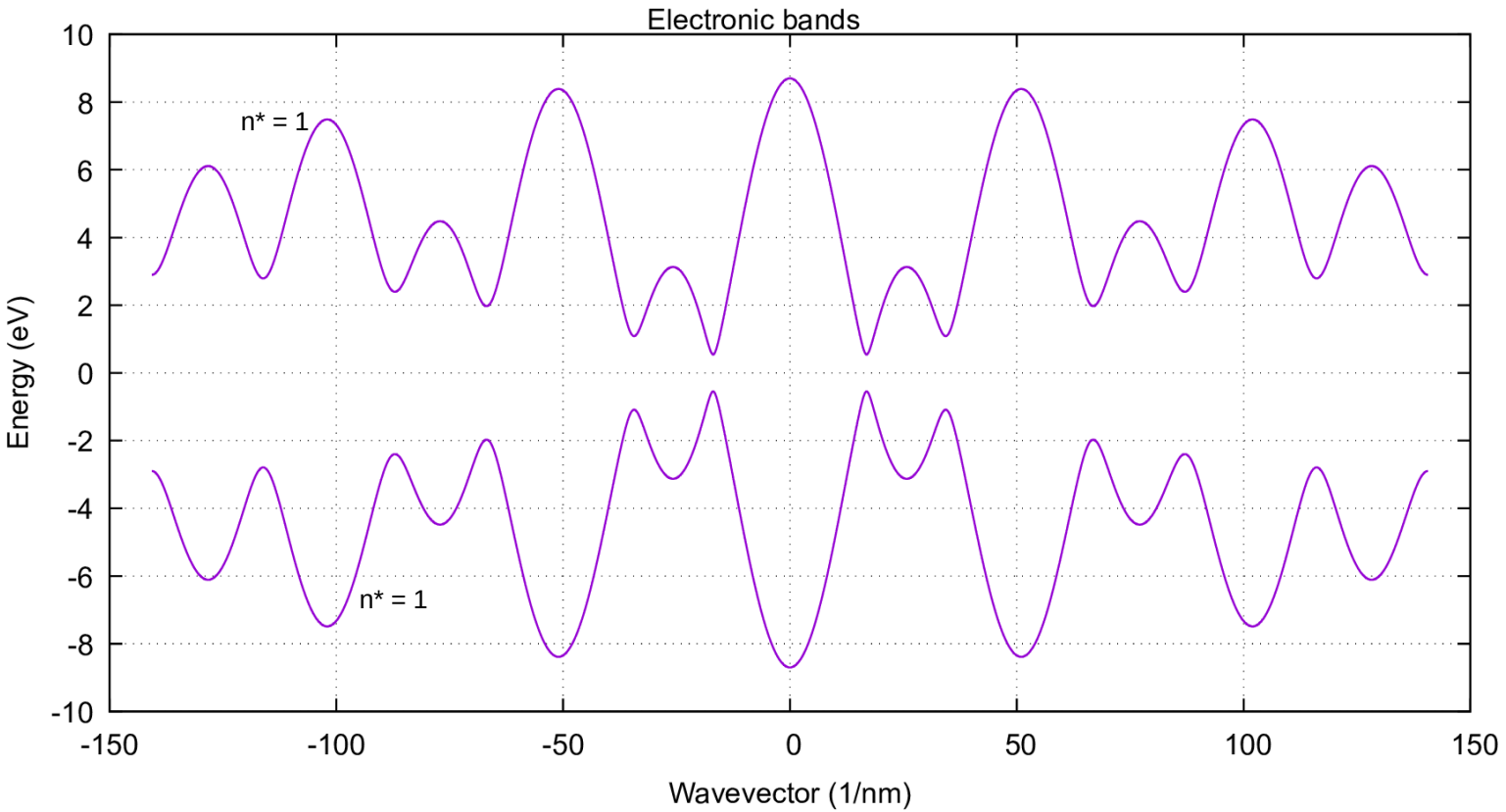
Tight binding electronic band structure of (6,5) CNTs



In reality, CNTs have a much more intricate band structure.

Helical electronic band structure of (6,5) CNTs





n* is the helical sub-band index, it is conserved!

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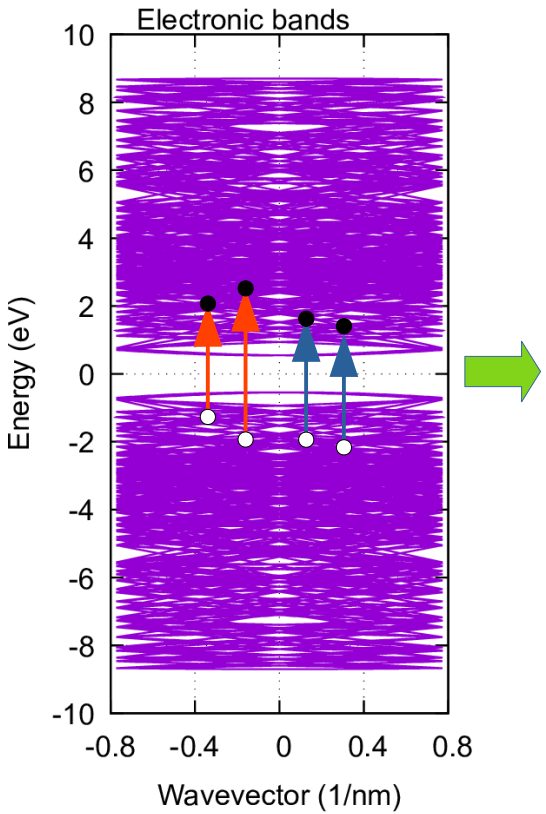
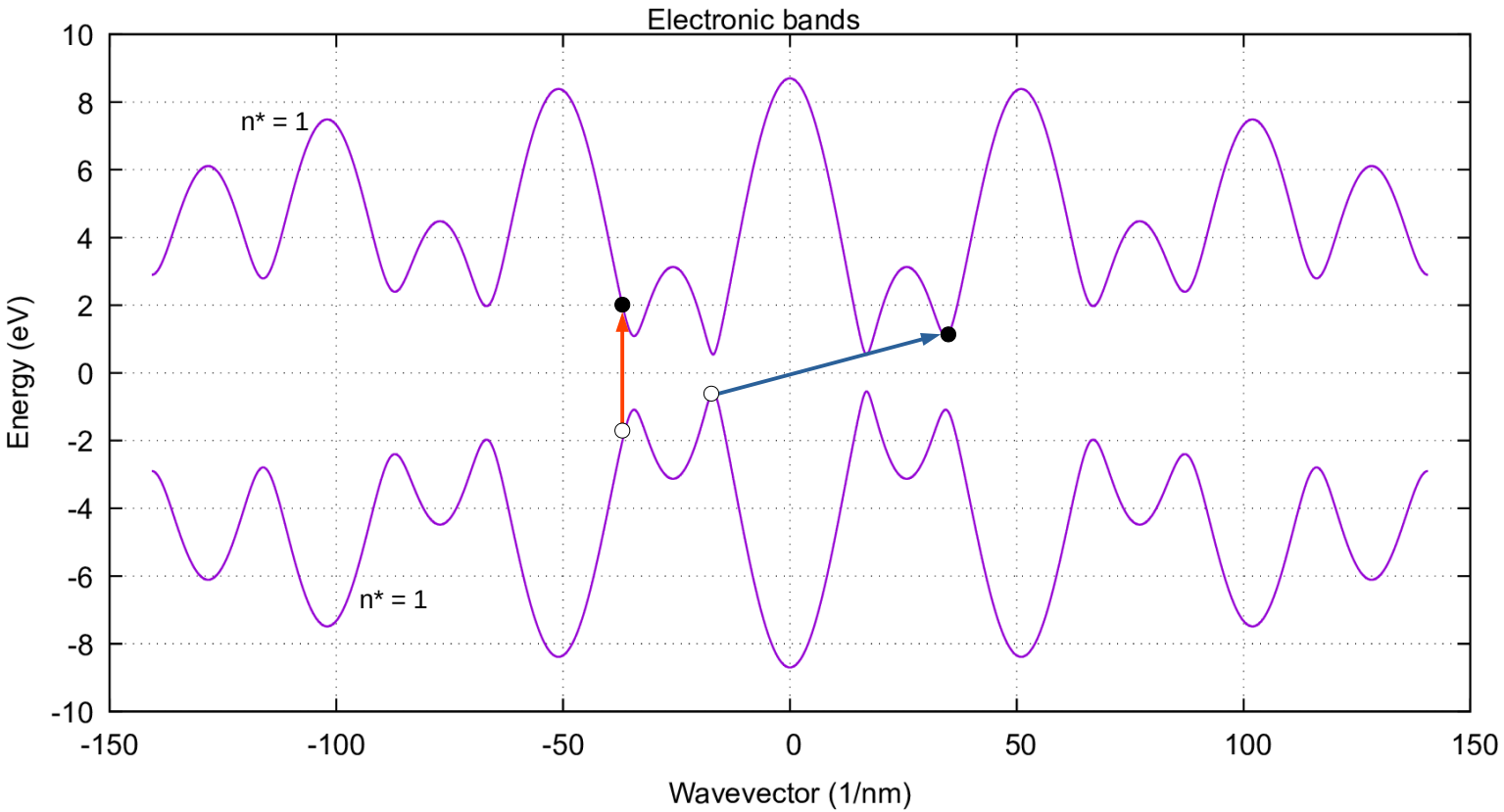
Helical electronic band structure of (6,5) CNTs





The helical wavevector is not conserved anymore!

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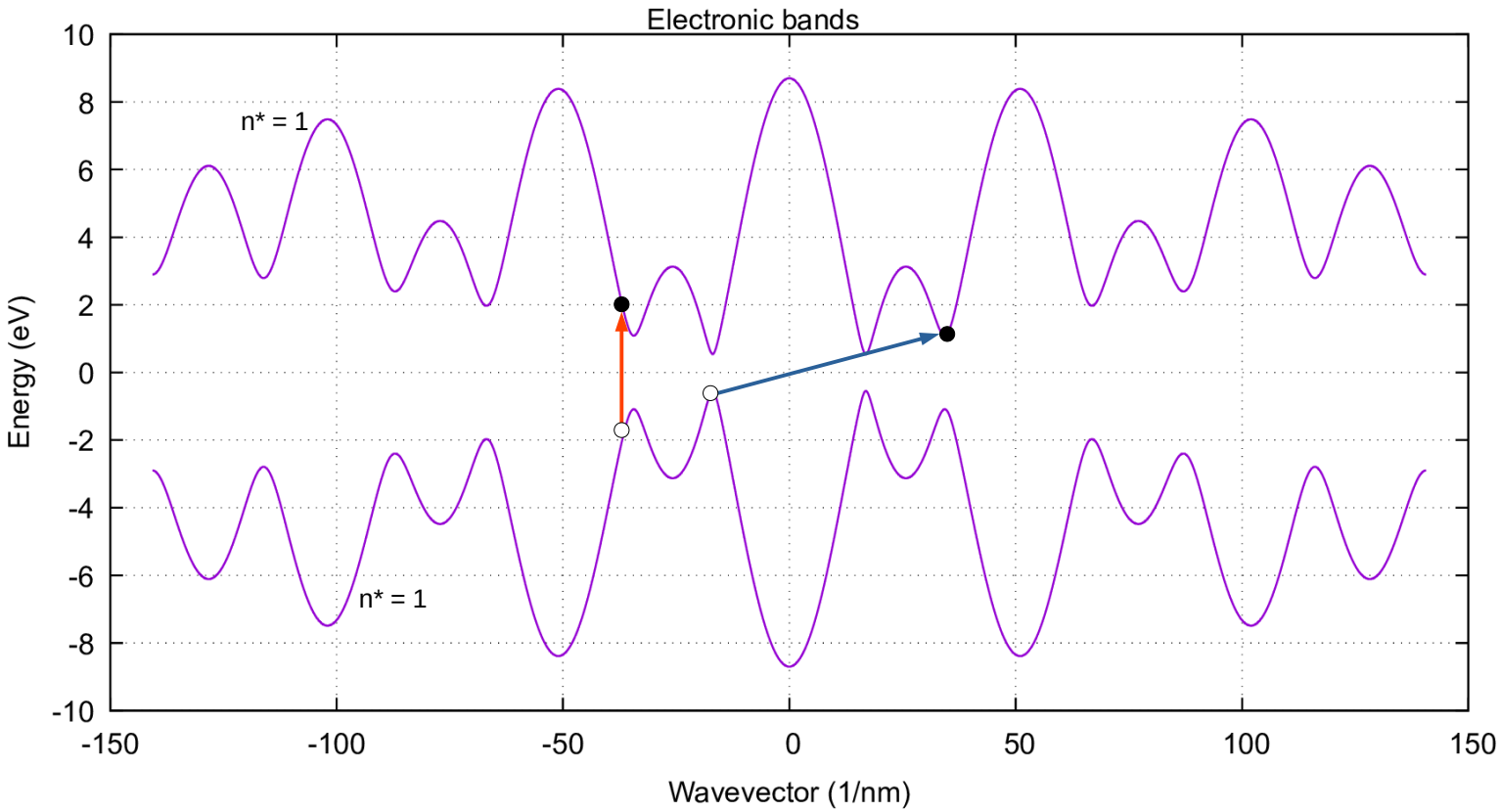
Helical electronic band structure of (6,5) CNTs

Using the helical representation gives us a number of advantages:

- easiness of computing the scattering integrals;
- possibility of including high-momentum carriers arising from phonon scattering;
- n* conservation;

This comes at the only cost of including a rigid k-shift when a transition crosses a sub-band.



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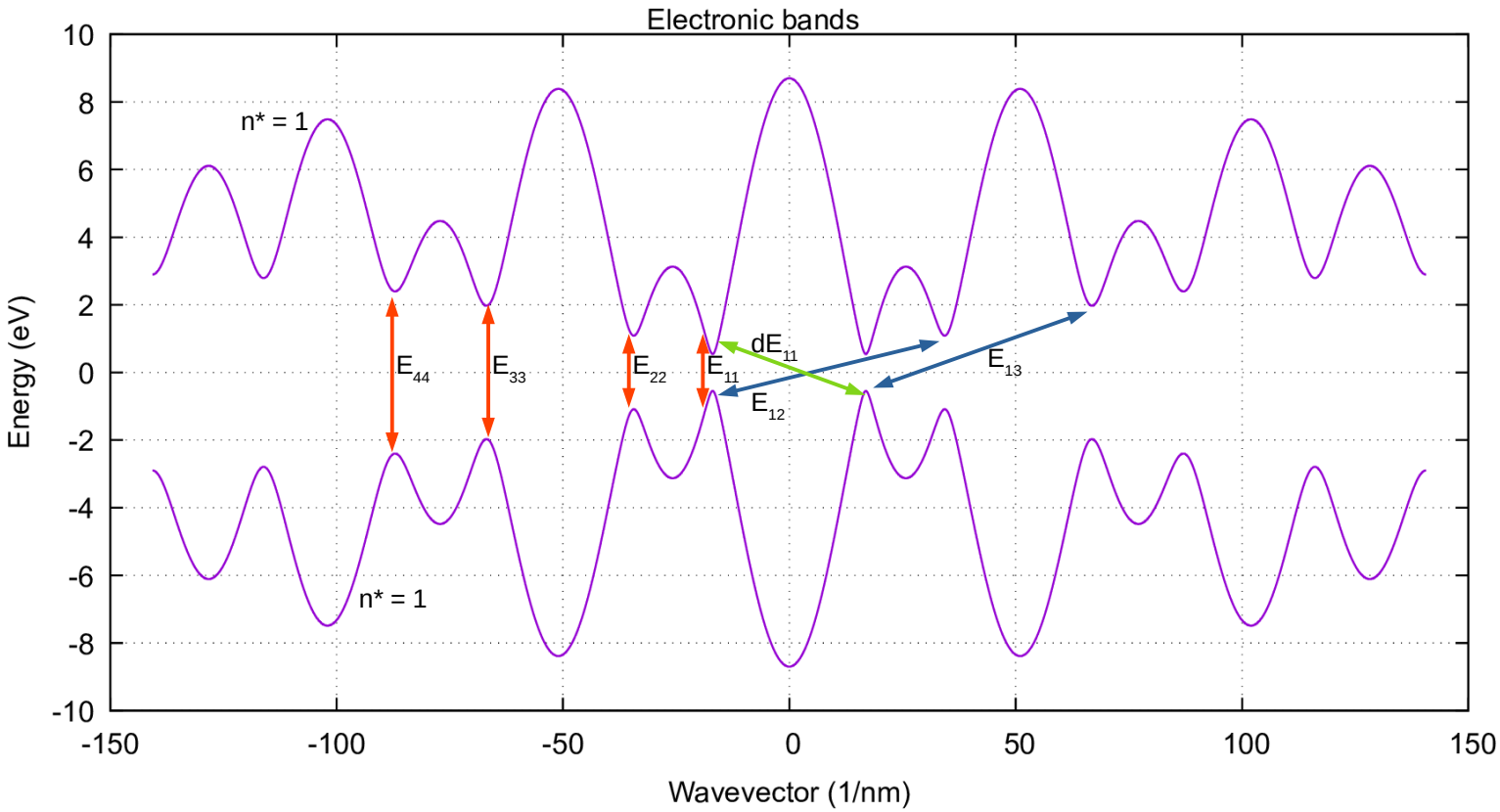
Helical electronic band structure of (6,5) CNTs

Moreover, we can label excitons just by looking at the extrema of the bands.

Colour code:

- Parallel polarised light excitons;
- Perpendicular polarised excitons;
- Dark excitons.



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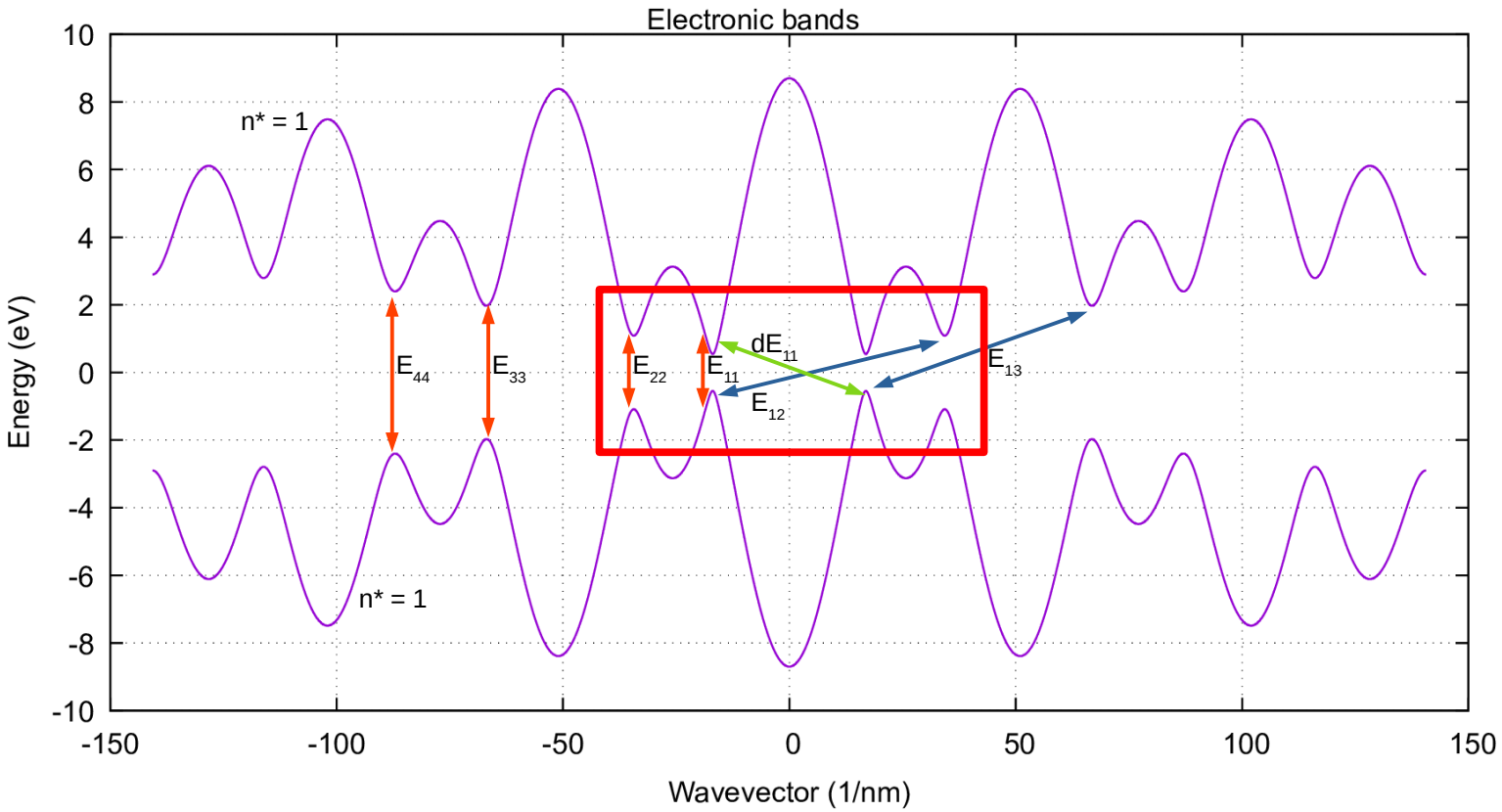
Helical electronic band structure of (6,5) CNTs

Moreover, we can label excitons just by looking at the extrema of the bands.

Colour code:

- Parallel polarised light excitons;
- Perpendicular polarised excitons;
- Dark excitons.

Finally, for the optical absorption we are interested only in this region.



Phonon band structure of (6,5) CNTs

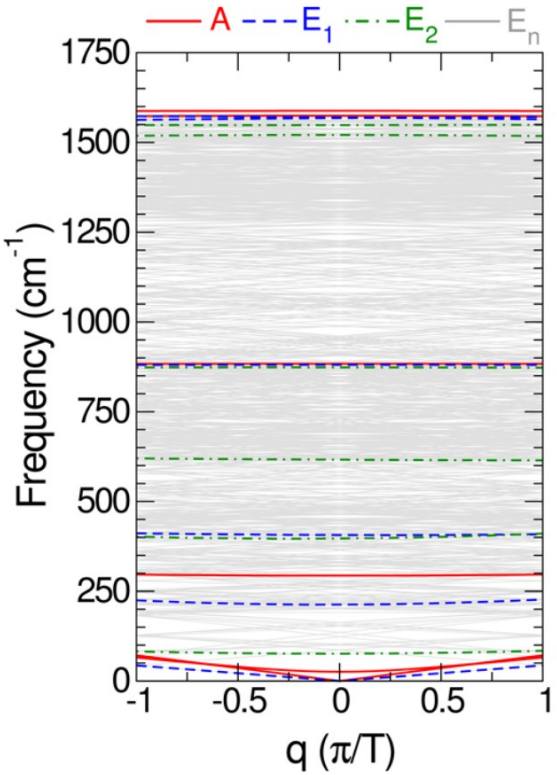
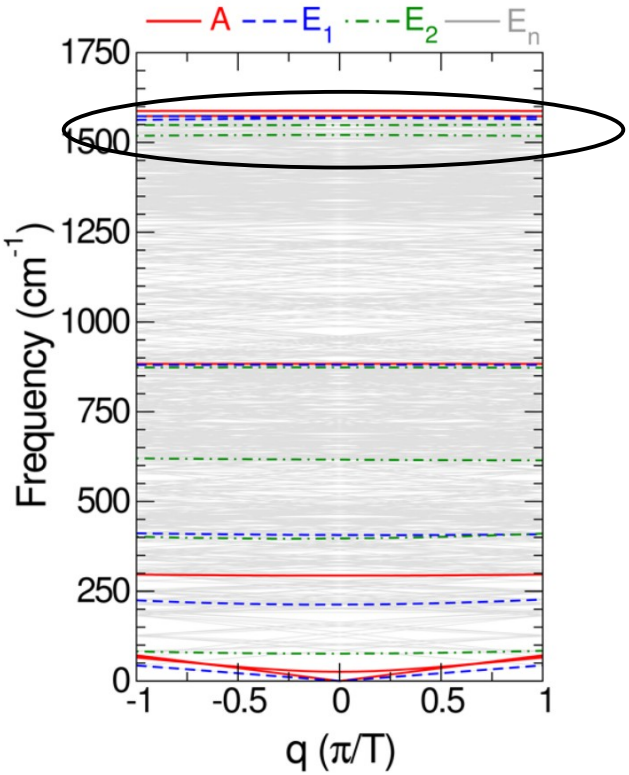


*Image taken from Lim et al., Nano Lett. 2014, 14, 1426−1432

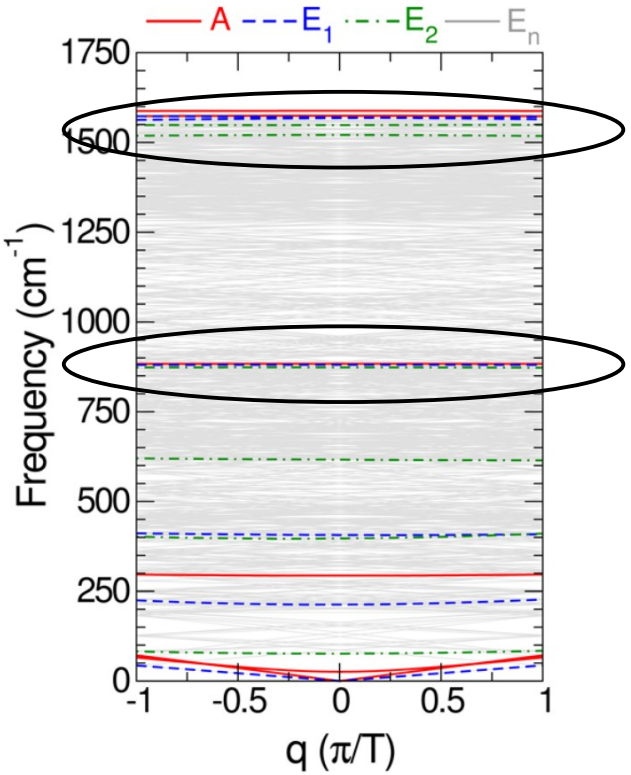
Phonon band structure of (6,5) CNTs



5th and 6th A (G phonon), E1 and E2 symmetries have similar energies

*Image taken from Lim et al., Nano Lett. 2014, 14, 1426−1432

Phonon band structure of (6,5) CNTs



5th and 6th A (G phonon), E1 and E2 symmetries have similar energies

4th A (oTO), E1 and E2 symmetries have almost identical energies

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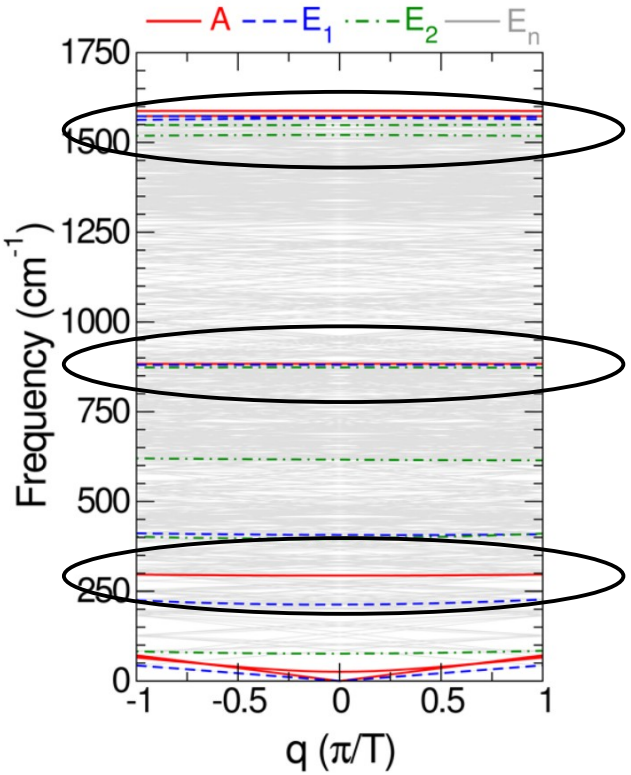
Phonon band structure of (6,5) CNTs



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RBM phonon

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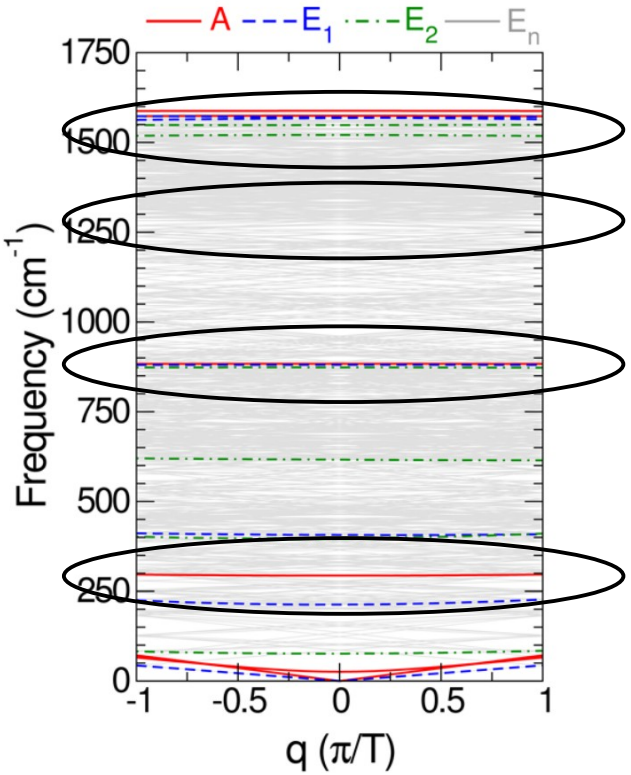
Phonon band structure of (6,5) CNTs



5th and 6th A (G phonon), E1 and E2 symmetries have similar energies

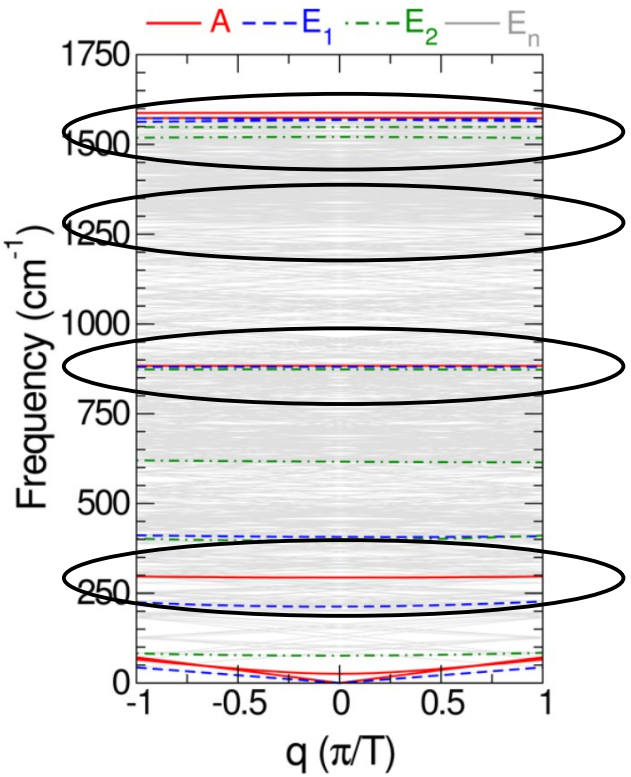
D phonon, somewhere here at 1312 cm^{-1}

4th A (oTO), E1 and E2 symmetries have almost identical energies

RBM phonon

*Image taken from Lim et al., Nano Lett. 2014, 14, 1426−1432

Helical phonon band structure of (6,5) CNTs



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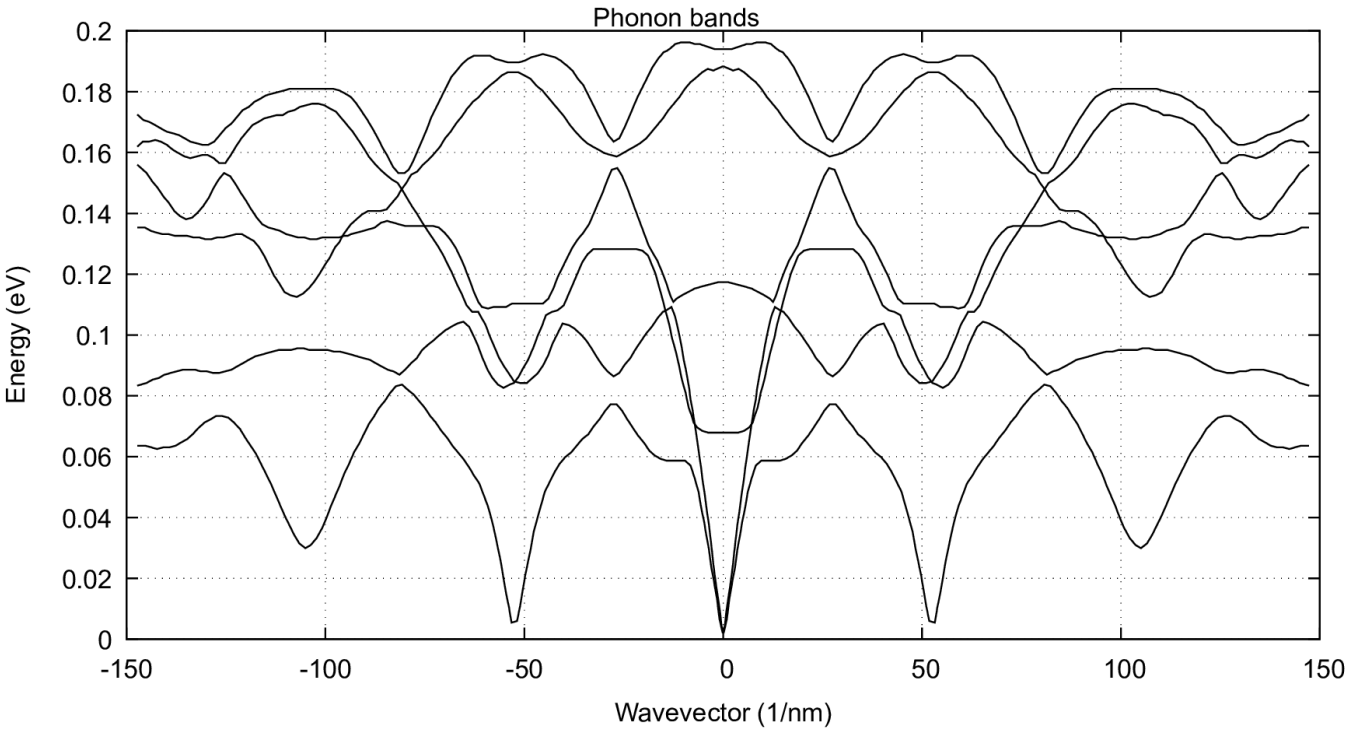
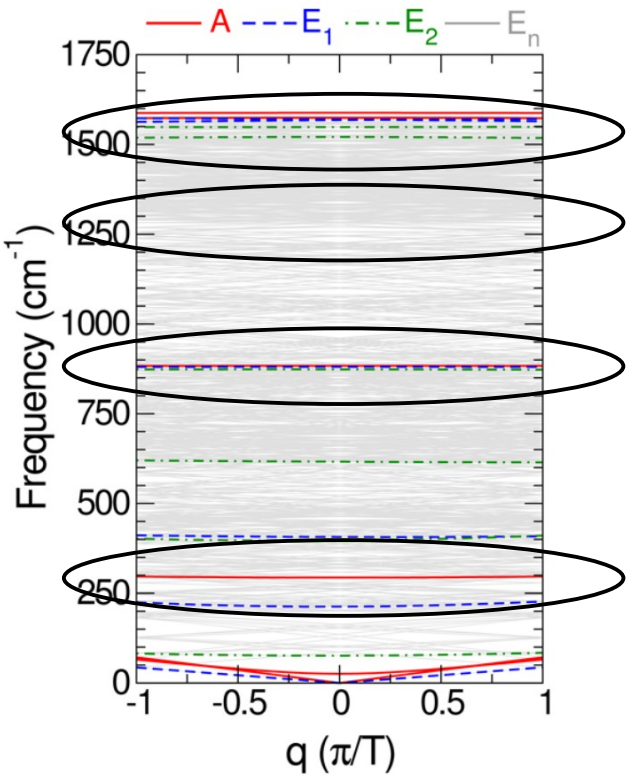


*image taken from Milosevic et al. Phys. Rev. B 72, 085426 (2005)

Helical phonon band structure of (6,5) CNTs



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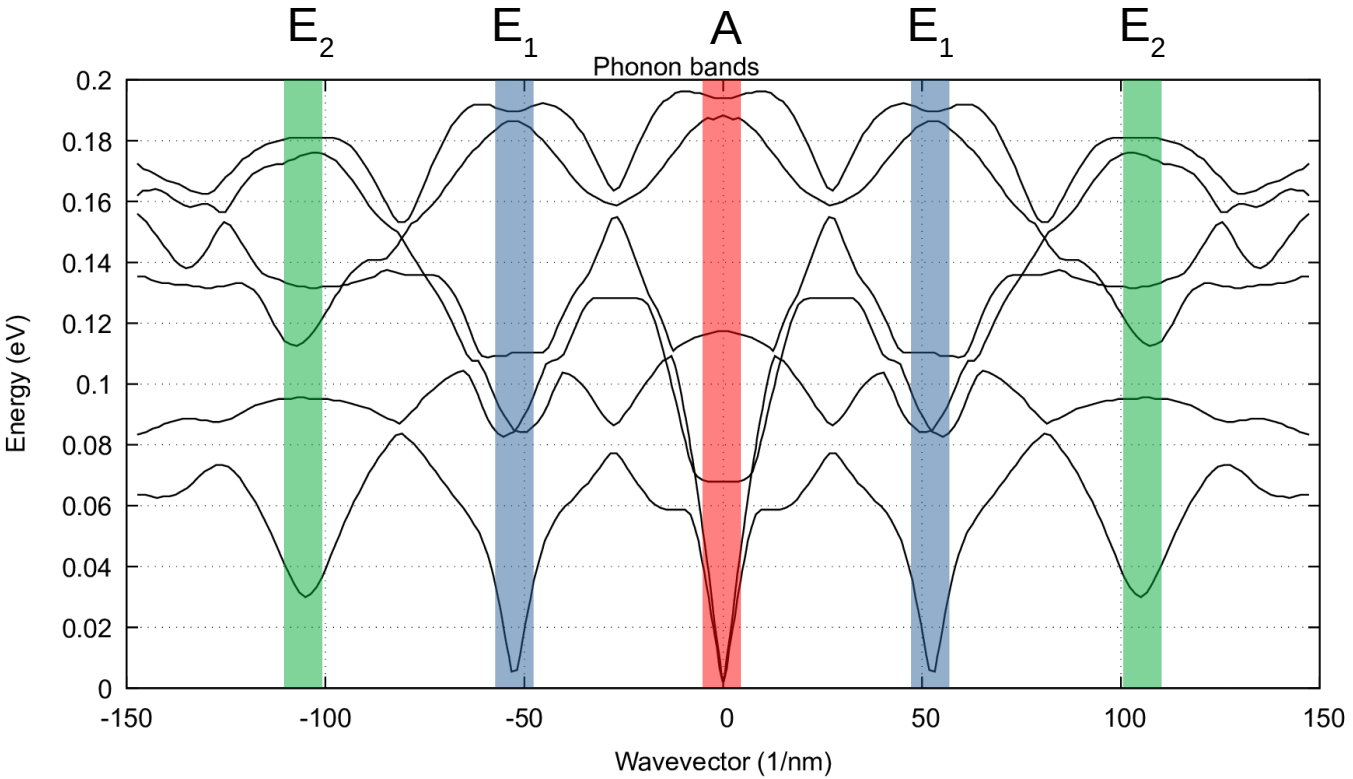

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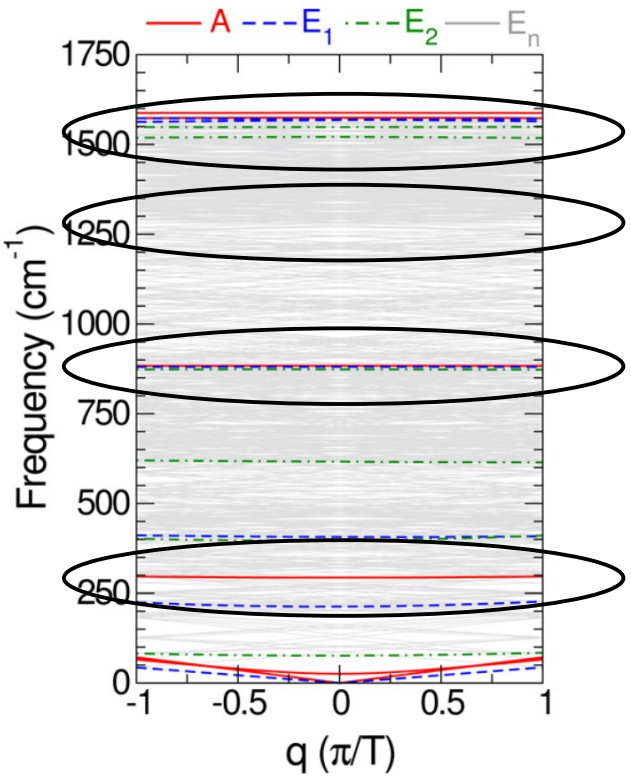
Helical phonon band structure of (6,5) CNTs



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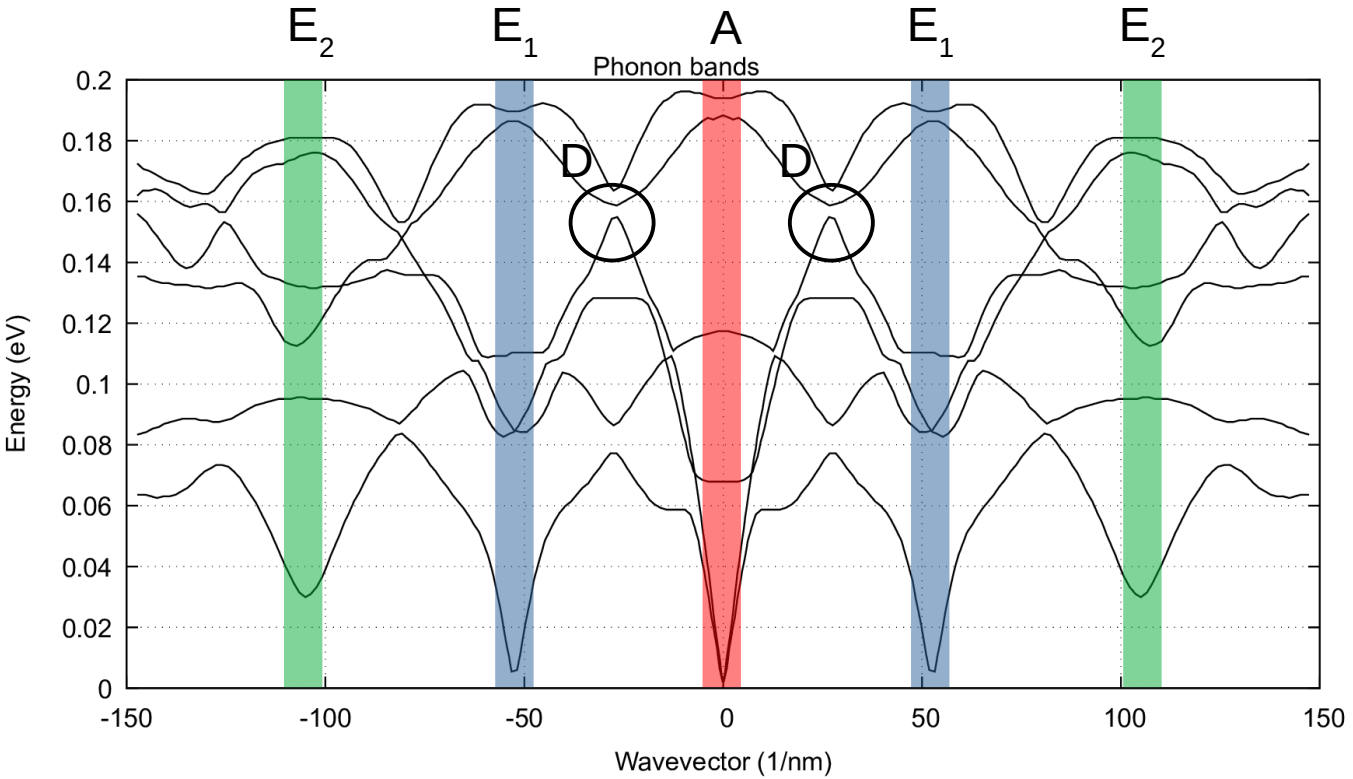

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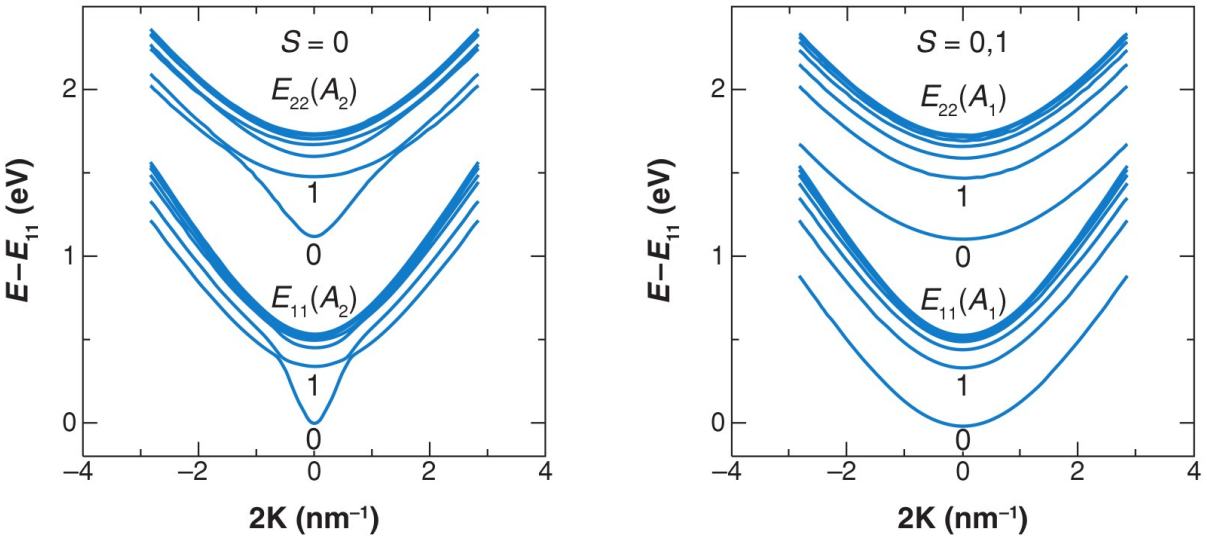
Exciton band structure of (6,5) CNTs



A symmetry exciton dispersion in (6,5) CNT.

*image from Mildred S. Dresselhaus, Gene Dresselhaus, Riichiro Saito, and Ado Jorio, Annu. Rev. Phys. Chem. 2007. 58:719-47